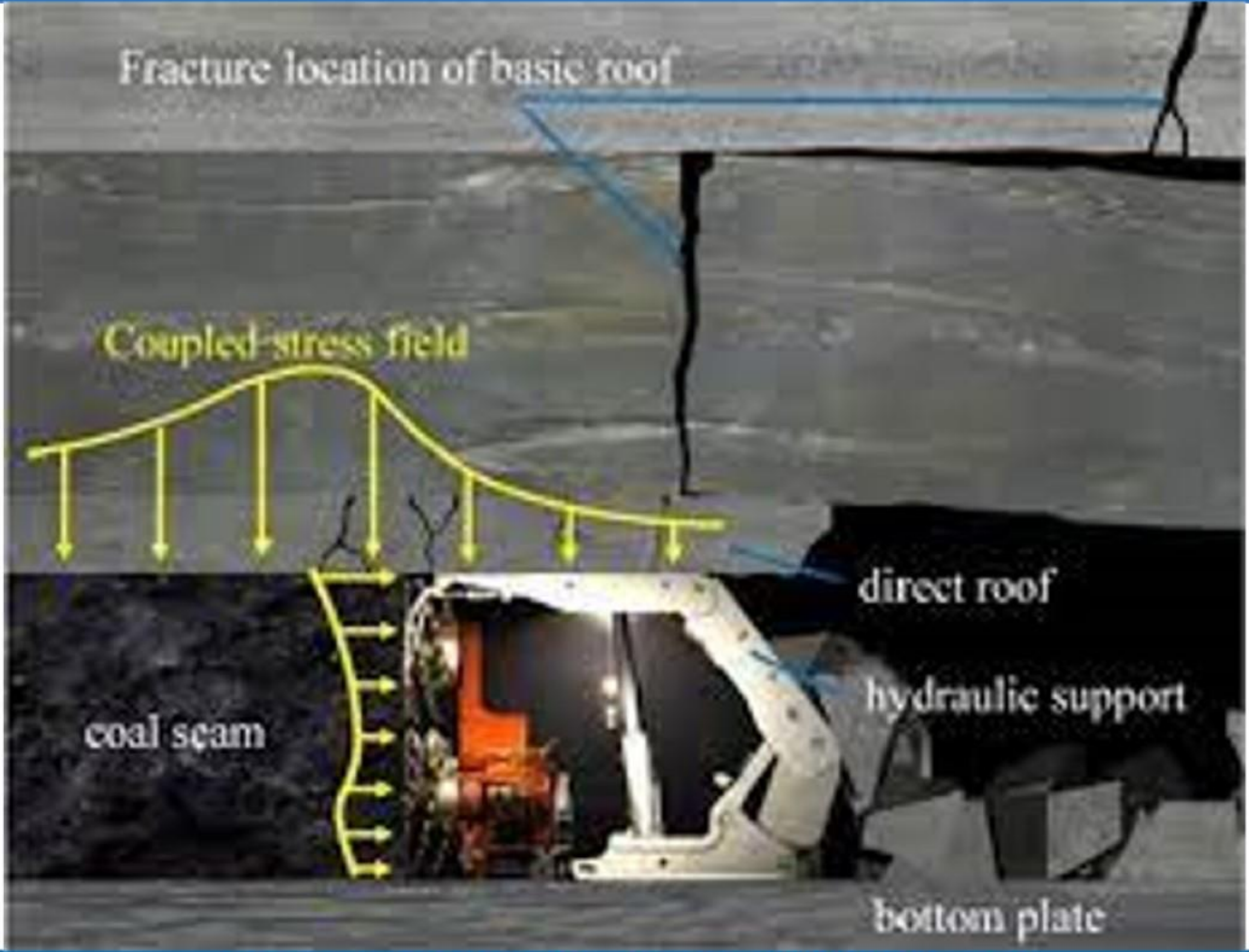
Syllabus

Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
2.	**Hard Roof Problems and Their Control with Case Study** ✓ Introduction ✓ Water Treatment Water injection for weakening Indian Research Hydraulic Fracturing ✓ Yield Pillar Technique Procedure for Design Yielding Chain Pillars ✓ Goaf Edge Blasting ✓ Goaf Edge Support Measures ✓ Rock Bursts Burst Proneness of coal and detection of bursts Drilling Yield Method Microseismic Method Rock Deformation Measurement Combating and Control of Bursts Volley Firing Hydraulic Fracturing Auger Drilling Design of Mine Layouts Burst Proneness of Indian Seams	5	**Students will learn about the hard roof problems and their control at deeper cover.**

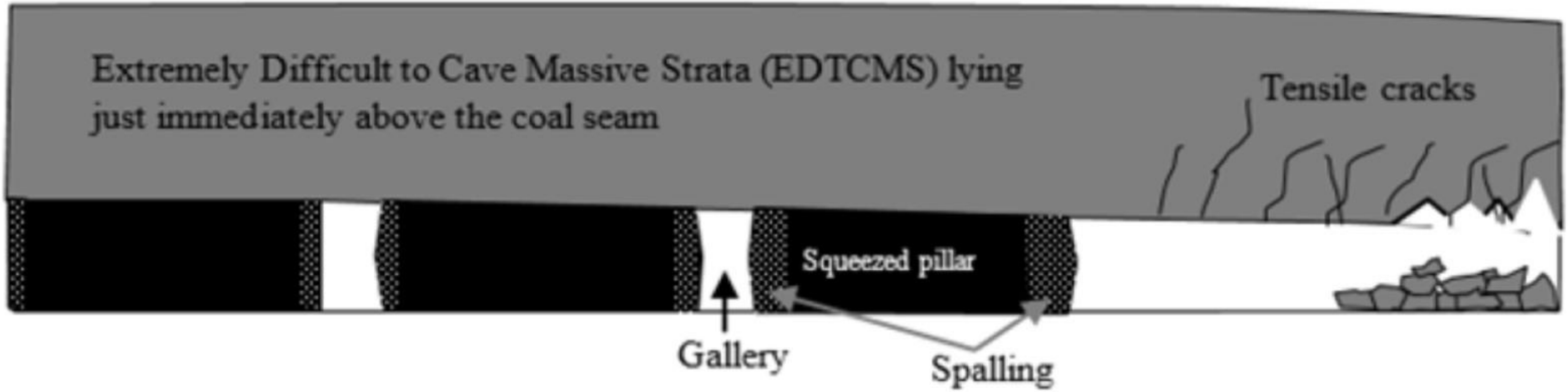
Longwall Face Section



Condition of Rib in Limestone Mine – Hourglass Failure

Load Transfer Pattern

SSP under Extremely Difficult to Cave Massive Strata (EDTCMS)



Caveability Index

On the basis of different field experiences, the caving characteristics of overlying strata are quantified in terms of Caveability Index (*I*) (Sarkar 1998), which is defined as:

$$I = \frac{\sigma l^{n} T^{0.5}}{5}$$

where, σ is uniaxial compressive strength in kg/cm^2, l is average length of core in cm, T is thickness of the strong bed in m and the factor *n* has a value of 1.2 in the case of uniformly massive rocks with a weighted average of RQD of 80% and above and in all other cases value of n is equal to 1.

Classification of roof	Caveability Index	Remarks
Easily caveable roof	I ≤2000	No perceptible weighting on face
Moderately caveable roof	2000 < I ≤ 5000	Weighting but not dynamic in nature
Roof caveable with difficulty	5000 < I ≤ 10,000	Weighting may or may not be dynamic in nature
Caveable with substantial difficulty	10000 < I ≤ 14000	Weighting may or may not be dynamic in nature
Caveable with extreme difficulty	I ≥ 14000	Weighting will be dynamic in nature

Introduction

- ✓ Unless stowing or filling is an available option, hard massive roof strata which are difficult to cave can cause serious difficulties in depillaring or longwalling.
- ✓ Such conditions can give rise to two problems:
- ✓ (i) Sudden caving of a part or whole of the roof over a large area causing sudden violent displacement of the enclosed air through available roadways or producing an air blast. This incident lasts for only a few seconds, producing violent compression and decompression of air and causing severe damage to men and materials.
- ✓ (ii) The second alternative, or additional hazard, can be a rock burst i.e. violent failure of rock ina roadway or at the working face. A rock burst is also variously termed as a coal bump, roof burst or floor burst, depending on which element of an excavation violently gives way.
- ✓ While an air blast can occur under any depth of cover, given the presence of hard strata, a rock burst can occur only at depths generally greater than 300 m and only if the coal itself is burst-prone.
- ✓ Also, an air blast can be completely prevented if stowing or filling is adopted.
- ✓ A rock burst, on the other hand, may occur in spite of filling, though its incidence may be reduced.

Introduction

- ✓ A rock burst may occur, again if the coal is burst-prone, due to an air blast, even if the latter takes place in an adjoining panel separated by competent pillars.
- ✓ Air blasts and rock bursts can obviously be clubbed together as violent manifestations of strain energy.
- ✓ Air blasts have been reported from several parts of the world. In India, they have occurred in different coalfields viz. Raniganj (Khottadih, Kuardi, Porascole, Dhemo Main collieries), Jharia (East Katras Project), Jhagrakhand (Churcha colliery), Singareni (Godavari Khani No. 8 and 10 Inclines).
- ✓ In the author's experience, an air blast can occur if the exposed goaf area is 4000 m^2 or greater for-working heights of up to 4.0 m, depending on the number of roadways available for the escape of air.
- ✓ Rock bursts in India are confined to a few coal mines of Raniganj coalfield, West Bengal.

Introduction

- ✓ Control of hard roof strata can be achieved by adopting the following methods in the absence of stowing or filling:
- ✓ 1. Water treatment of the strata
- ✓ 2. Yield pillar technique
- ✓ 3. Roof blasting
- ✓ 4. Goaf edge support measures
- ✓ These methods are discussed below primarily with air blasts in view, through the first two may have application for rock burst control.

WATER TREATMENT

- ✓ Hard strata can be treated with water in two ways:
- ✓ 1. Water injection under pressure to weaken the rock matrix as well as joints and bedding planes.
- ✓ 2. Hydro-fracturing to create fractures in the directions which are amenable to caving.
- ✓ Both techniques have been primarily designed for longwall extraction but can be considered equally effective for depillaring of bord and pillar workings.
- ✓ The first technique is for preconditioning the rock mass before commencing extraction.
- ✓ Some sandstones, which are fine-grained or have low clay content and are particularly massive, may not weaken sufficiently to improve cavability with this technique.
- ✓ It has then to be coupled with blasting or else the second technique of hydro-fracturing may have to be adopted.

WATER TREATMENT

Water injection for weakening

- ✓ This technique is being practised in Chinese coalfields since the late 1970's (Xu Linsheng, 1987; Chen Hui and Niu Xizhuo 1987; Pan et al., 1983) where goaf fall areas up to 151,000 m^2 in bord and pillar extraction at Datang mine under massive sandstones and conglomerate have been reported, such large areas existing because of pillar remnants.
- ✓ Switching to longwall coupled with water injection, successful extraction has been achieved.
- ✓ Also reported by Xu Linshemg are the experiences at several other mines.
- ✓ The technique consists of injecting water at the rate of 30-60 litres/min at a pressure of 15.3-30.6 MPa, using a pumping arrangement shown schematically in **Figure 10.1**, through holes inclined over the (planned) goaf area.
- ✓ The holes are spaced 30 m apart and are 60-90 mm in diameter.
- ✓ They are drilled at a low angle of 17°-24° with the horizontal and have a length sufficient to cover a sizable roof thickness of an entire hard bed as the case may be.
- ✓ The holes are also inclined at about 70° towards the start of the face.

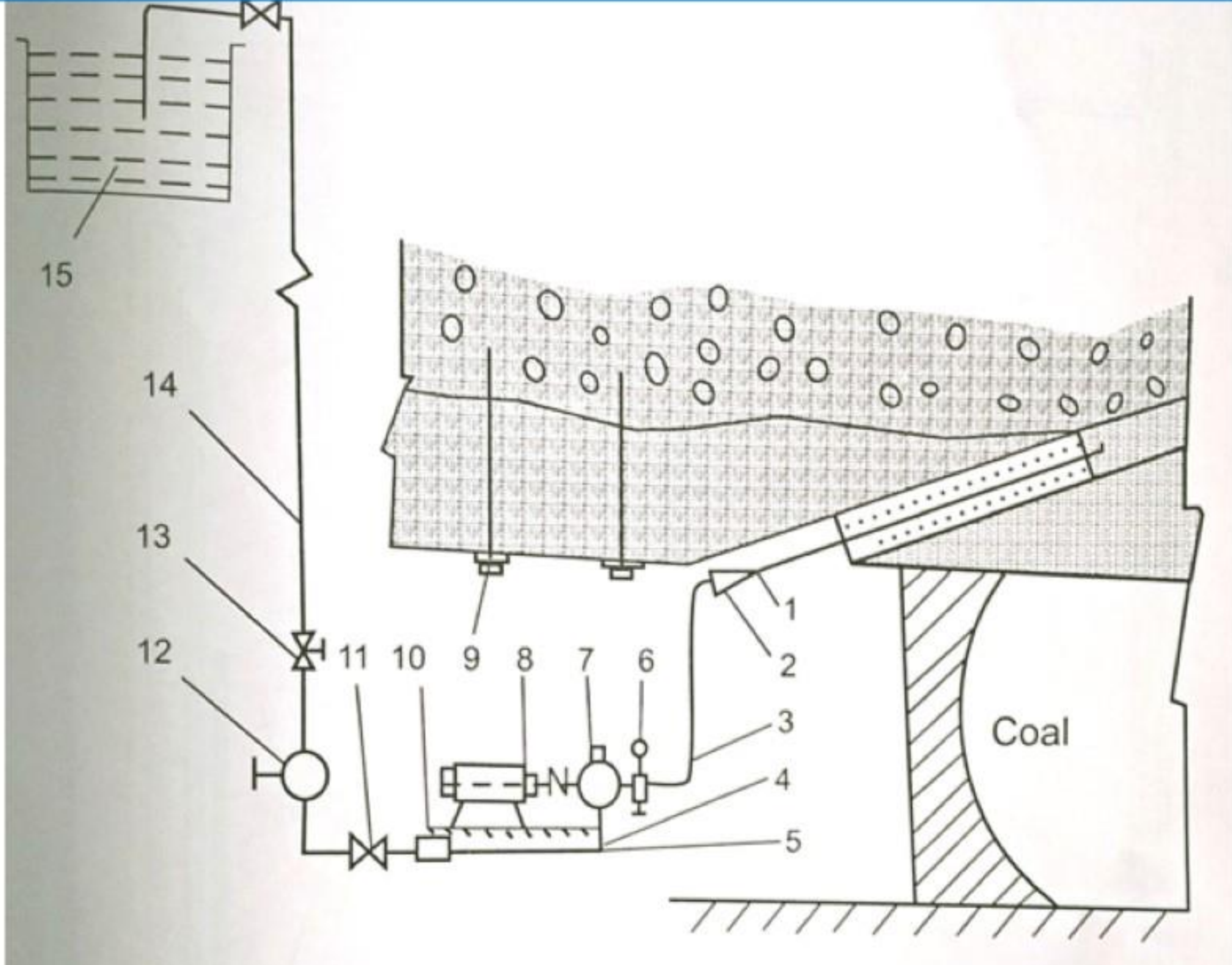


Fig. 10.1. 1.High-pressure water jetting system; 2. Pipejoint; 3. High-pressure rubber pipe; 4. Pedestal; 5. Low-pressure rubber pipe; 6. Manometer; 7. High pressure pump; 8. Electric motor; 9. Bolt; 10. Flowmeter; 11. Distribution valve; 12. Filter; 13. Valve; 14. Water pipe; 15. Cistern on surface (after Xu Linsheng, 1987).

Water injection for weakening

- ✓ The holes are drilled from one or both gate roads depending on the face length.
- ✓ Typical drilling patterns from two mines are shown in **Figures 10.2 & 10.3**.
- ✓ The mouth of each hole is sealed with an inflatable rubber packer or cement mortar and water is retained for several days, continuously pumping it as required.
- ✓ Separate holes are recommended for two or more hard strata if present, adopting different lengths of the mortar seal.
- ✓ Water injection is also recommended for treating standing goaf overhangs.

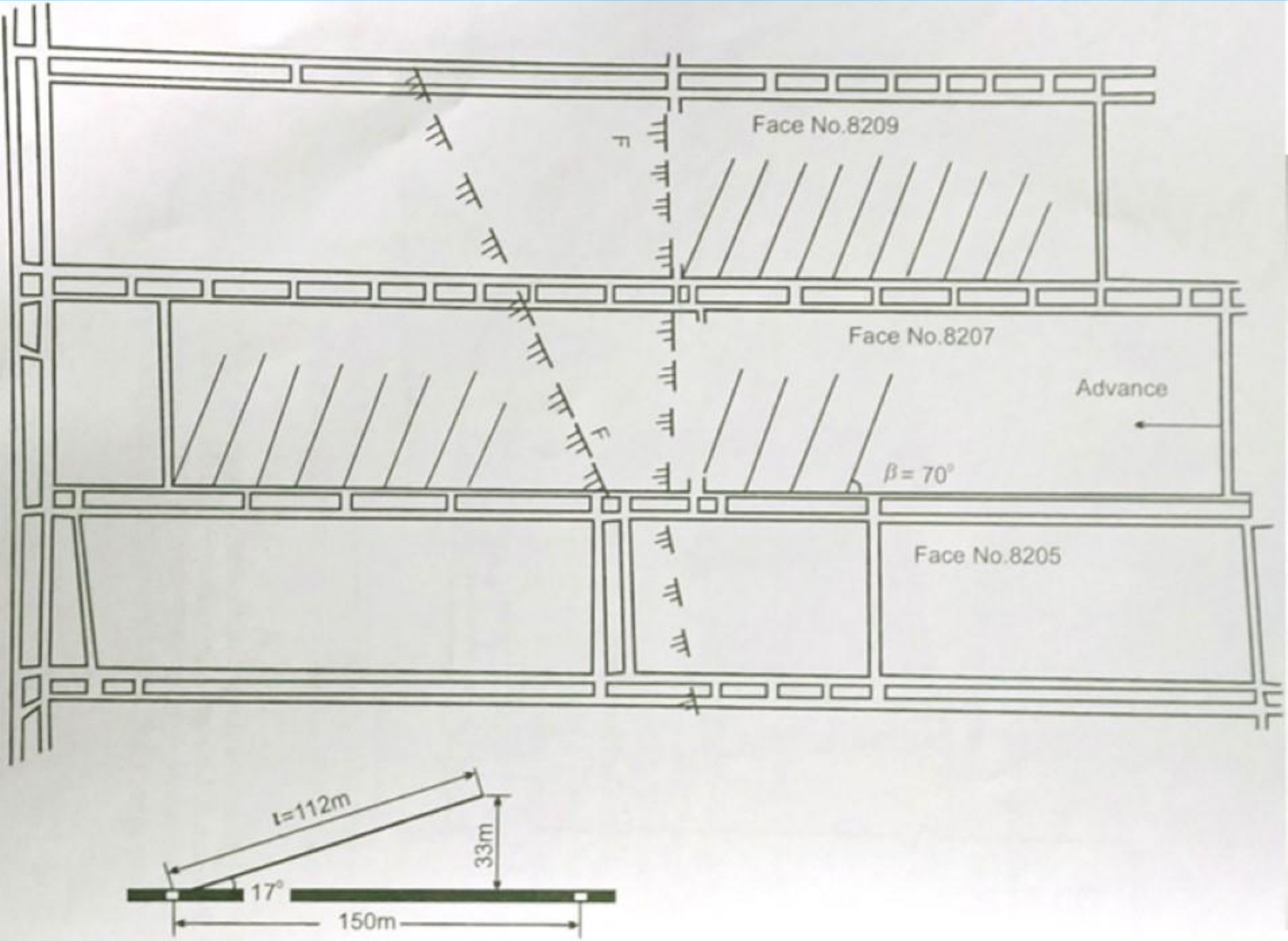


Fig. 10.2. Pattern of drilling for water injection of Silaogou mine (no drilling in faulted zone F-F) (after Xu Linsheng, 1987).

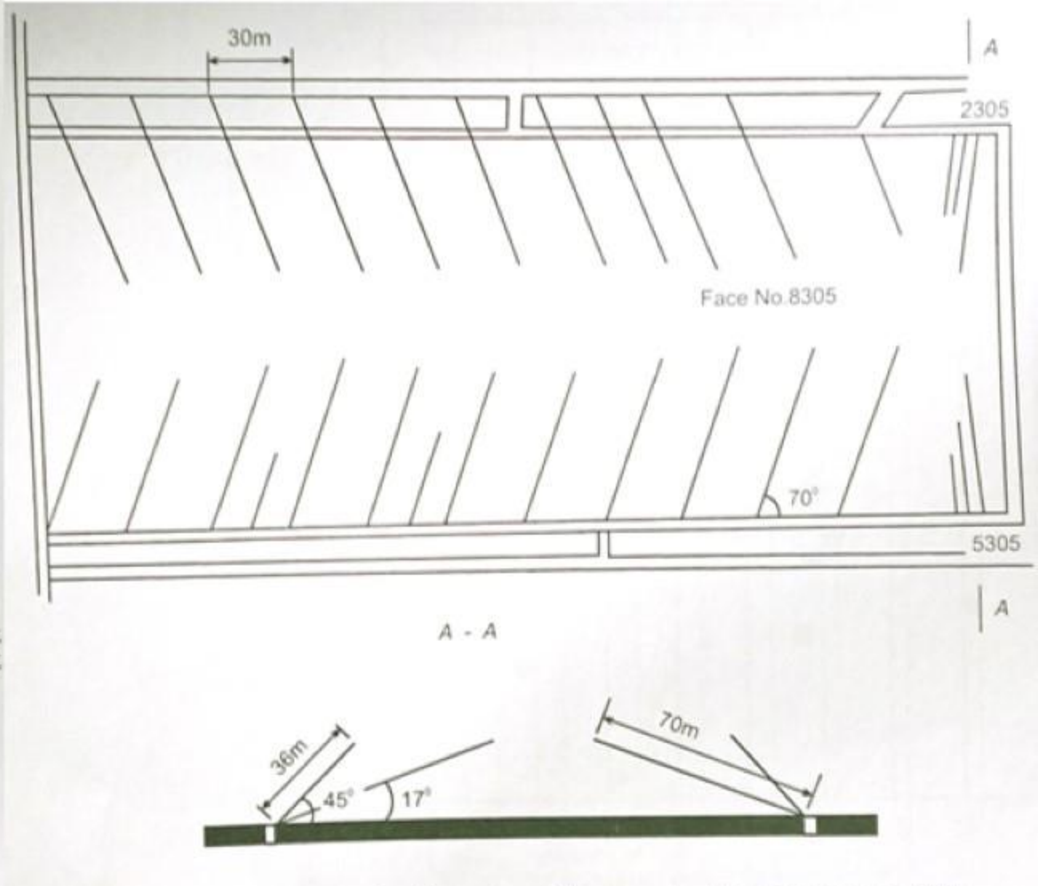


Fig. 10.3. Drilling pattern for a longer face of Yungang mine (after Xu Linsheng, 1987).

WATER TREATMENT

Indian Research

- ✓ Field trails of water injection as done in China have been conducted in two bord and pillar mines (CMRI Report, 2000):
  - ✓ Parascole West of Eastern Coalfields Limited, Raniganj coalfield
  - ✓ Churcha collicry of South Eastern Coalfields Limited, Jhagrakhand coalfield
- ✓ Initial laboratory experiments indicated that the strength reduction after seven days of water immersion depends on clay content as well as grain size of sandstone, **Figures 10.4 & 10.5**.
- ✓ The result in these figures indicates that sandstone with fine grains or low clay content may not show improved caving unless at least a few joints and bedding planes exist.
- ✓ Parascole West with a massive sandstone of over 20 m thickness was practising yield pillar method for depillaring at the time of the water injection trial.
- ✓ The yield pillars took several months to yield and allow the roof to cave (Sheorey ef al., 1995).
- ✓ The water injection experiment was therefore not very conclusive.

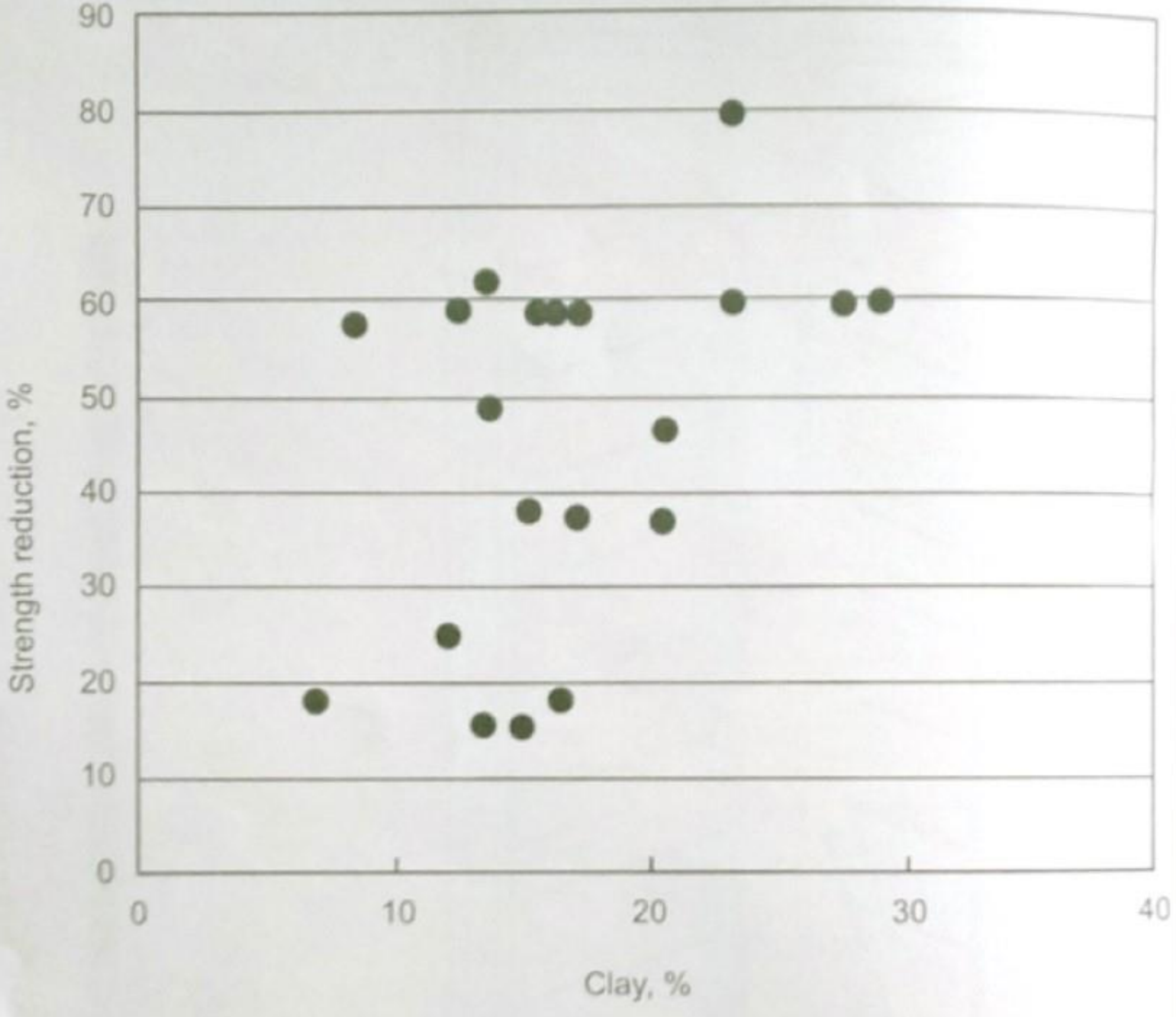


Fig. 10.4. Compressive strength of sandstones vs. clay content after 7 days of water immersion.

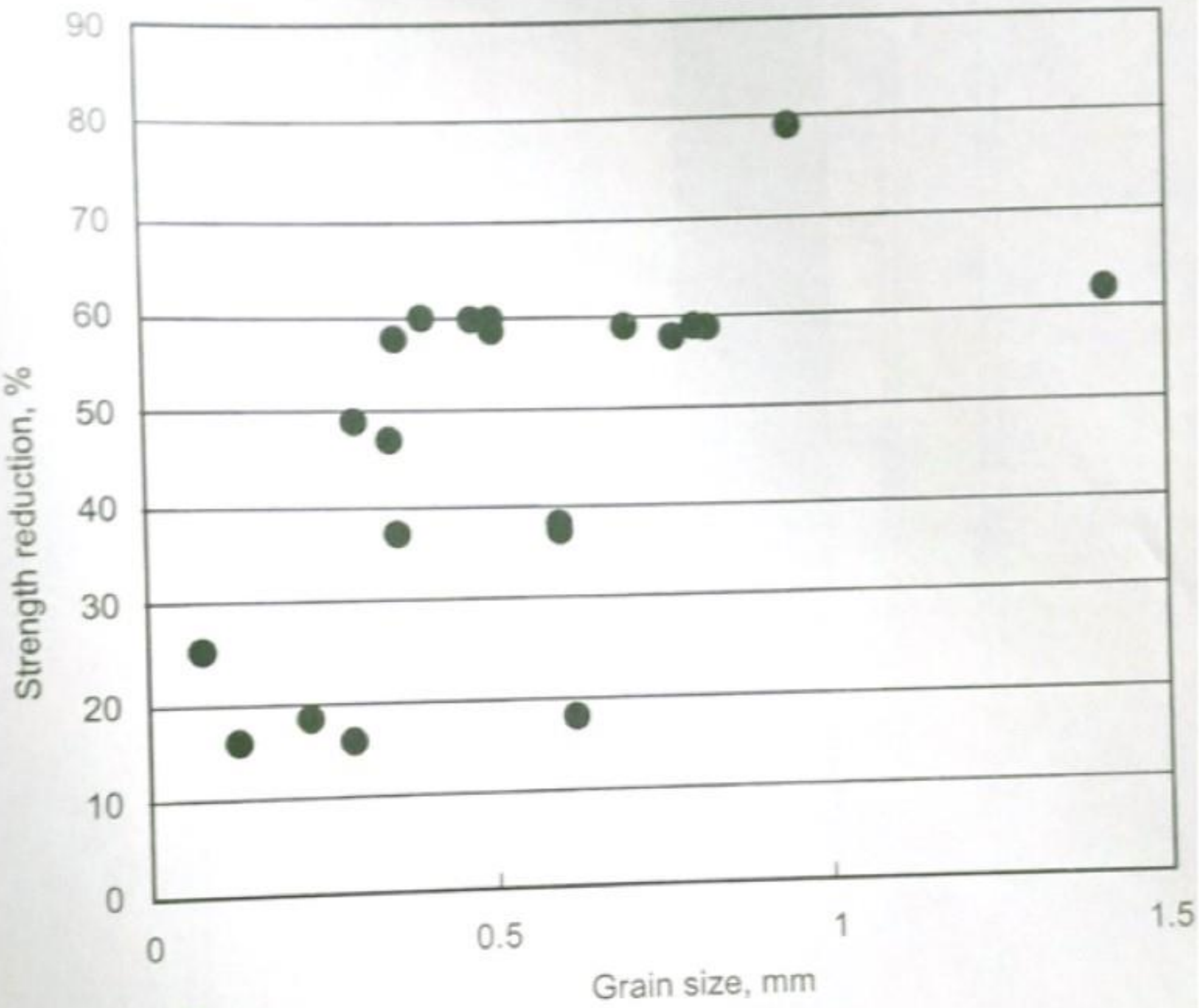


Fig. 10.5. Compressive strength of sandstones vs. grain size after 7 days of water immersion.

WATER TREATMENT

Indian Research

- ✓ At Churcha West the field trial gave more useful results (Banerjee et al., 2003).
- ✓ A borehole section at this mine is shown in **Figure 10.6** and a plan of the experimental panel is given in **Figure 10.7** showing the location of a total of 11 injection holes.
- ✓ The holes were 54 mm in diameter and were 65-70 m long, inclined at 25°-30° towards the approaching goaf.
- ✓ The intact rock compressive strength, which was 40-50 MPa, reduced after 7 days of water immersion by about 30-60% depending on the sandstone horizon.
- ✓ The holes were sealed with cement mortar over a length of 3-5 m and the initial water pressure was 1400-2400 psi (9.6-16.5 MPa), which was somewhat lower than the Chinese values.
- ✓ The flow was maintained at 40-68 litres/min for 4 to 9 hrs.
- ✓ A total of 15000 to 32000 litres were injected in each hole.
- ✓ The pressure was found to fall for some time before reaching a stabilized limit.

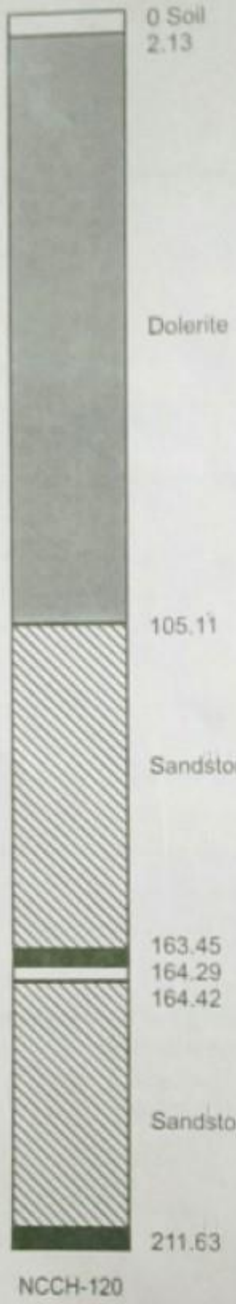


Fig. 10.6. Borehole section of borehole No. NCCH-120 at Churcha (after Banerjee *et al.*, 2003).

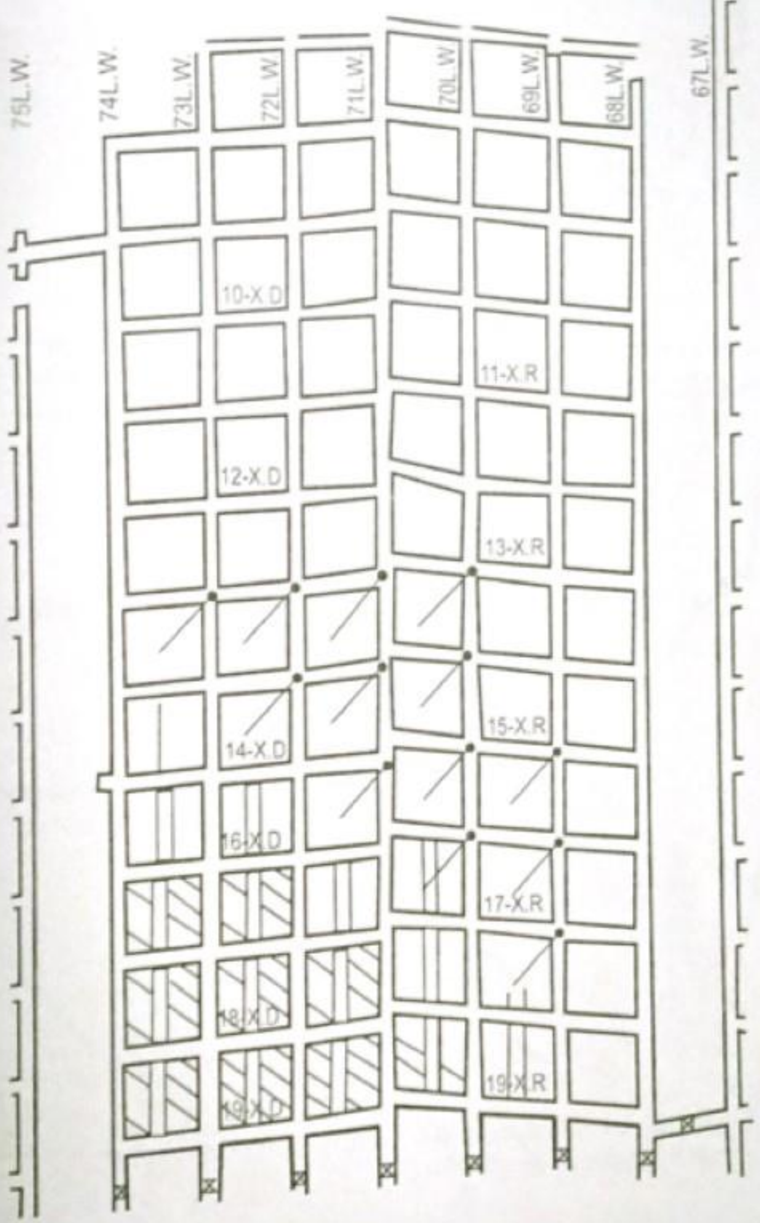


Fig. 10.7. Location of holes drilled for water injection in 10W extended panel (after Banerjee *et al.*, 2003).

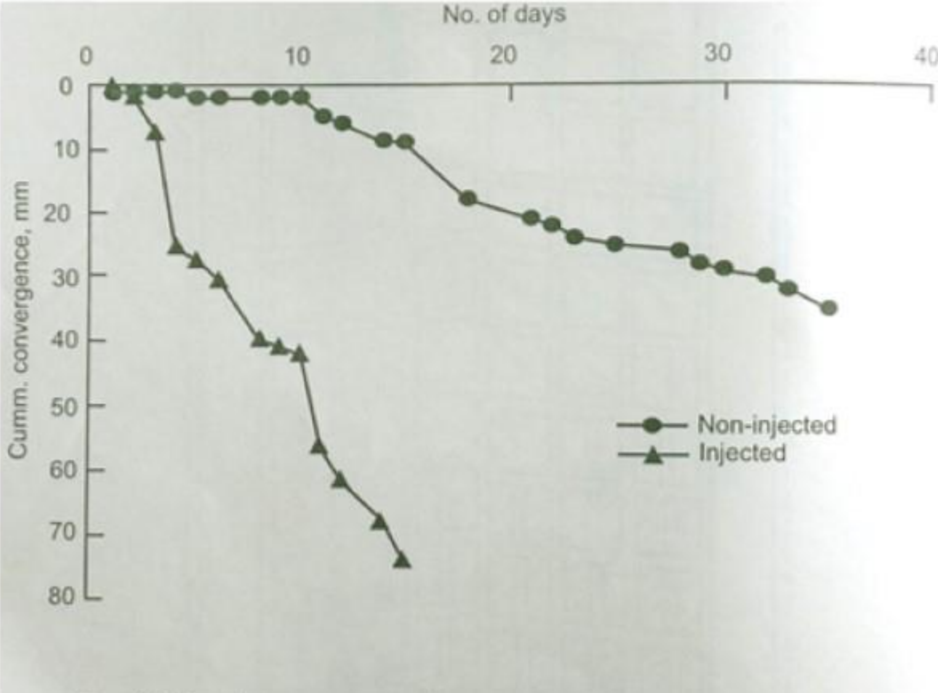


Fig. 10.8. Cumulative convergence in non-injected and water injected zones (after Banerjee *et al.*, 2003).

WATER TREATMENT

Indian Research

- ✓ The following observations were made:
  - ✓ The roof-to-floor convergence in the injected and non-injected areas was widely different, as typified by Figure.
  - ✓ Some symptoms of roadway roof instability like minor falls, widening of joints, tensile cracks, water percolation were noticed.
  - ✓ Pillar spalling in injected zones was significantly less.
  - ✓ Caving in the goaf produced much smaller fragments and caving was accompanied by water.
  - ✓ Water injection was regarded as overall successful in the bord and pillar extraction at Churcha.

Hydraulic Fracturing

- ✓ Successful experiments in block caving in a copper-gold mine (van As and Jeffrey, 2000) and in caving the standing goaf overhang of a coal mine (Mills et al., 2000) have been reported from Australia using hydrofrac equipment, which is generally used for insitu stress measurement.
- ✓ Australian experience in a mine indicated that water injection before extraction did not necessarily induce regular caving and some danger of air blasts remained.
- ✓ Hydraulic fracturing was therefore thought of as an alternative to induce caving and is thus specific to overhanging goafs.
- ✓ Since hydraulic fracturing causes the fracture to develop in a borehole at right angles to the minimum principal stress, it is necessary to know beforehand the principal stress directions.
- ✓ These directions may not be suitable before extraction but may become so after extraction, to induce proper caving.
- ✓ Thus insitu stress measurement as well as numerical modelling are required.

WATER TREATMENT

Hydraulic Fracturing

- ✓ At Moonee colliery in Australia, where this technique was successfully tried (Mills et al., 2000), longwalling was being practiced below a 30-35 m thick hard conglomerate bed prone to air blasts with goaf falls of 5,000-30,000 m^2.
- ✓ The rock had a compressive strength of 50-80 MPa and tensile strength of 4 MPa.
- ✓ This technique was routinely applied after the first successful trial about once in 10 days to induce regular caving at this mine, eliminating the fear of air blasts. Mills et al. have not, however, clearly specificd the direction and disposition of holes.

YIELD PILLAR TECHNIQUE

- ✓ The concept of using post-failure pillar behaviour in bord and pillar design, particularly at greater depths, was put forward by Starfield and Fairhurst (1968).
- ✓ Pillar failure in a bord and pillar panel surrounded by wide, strong chain pillars can be catastrophic (unstable) or progressive (stable) depending on strata stiffness and proximity of the chain pillars.
- ✓ The behaviour will thus be akin to that of a rock specimen loaded in a stiff machine.
- ✓ Salamon (1970) has given a complete mathematical treatment for a panel of pillars (which may be of different size).
- ✓ The concept put forward by Starfield and Fairhurst is however much simpler to understand.
- ✓ If a pillar in a panel is replaced by a hydraulic jack as in **Figure 10.9**, which is slowly retracted, the initial force on it due to rock load will drop linearly with a corresponding increase in roof lowering of roof-to-floor convergence, **Figure 10.9b**.
- ✓ The slope of this characteristic, which is termed the local mine stiffness, will be different for different pillar locations, depending on proximity of the stable chain pillars.
- ✓ We now replace the jack by the pillar itself and superpose the local stiffness on the post-failure characteristic of this pillar.

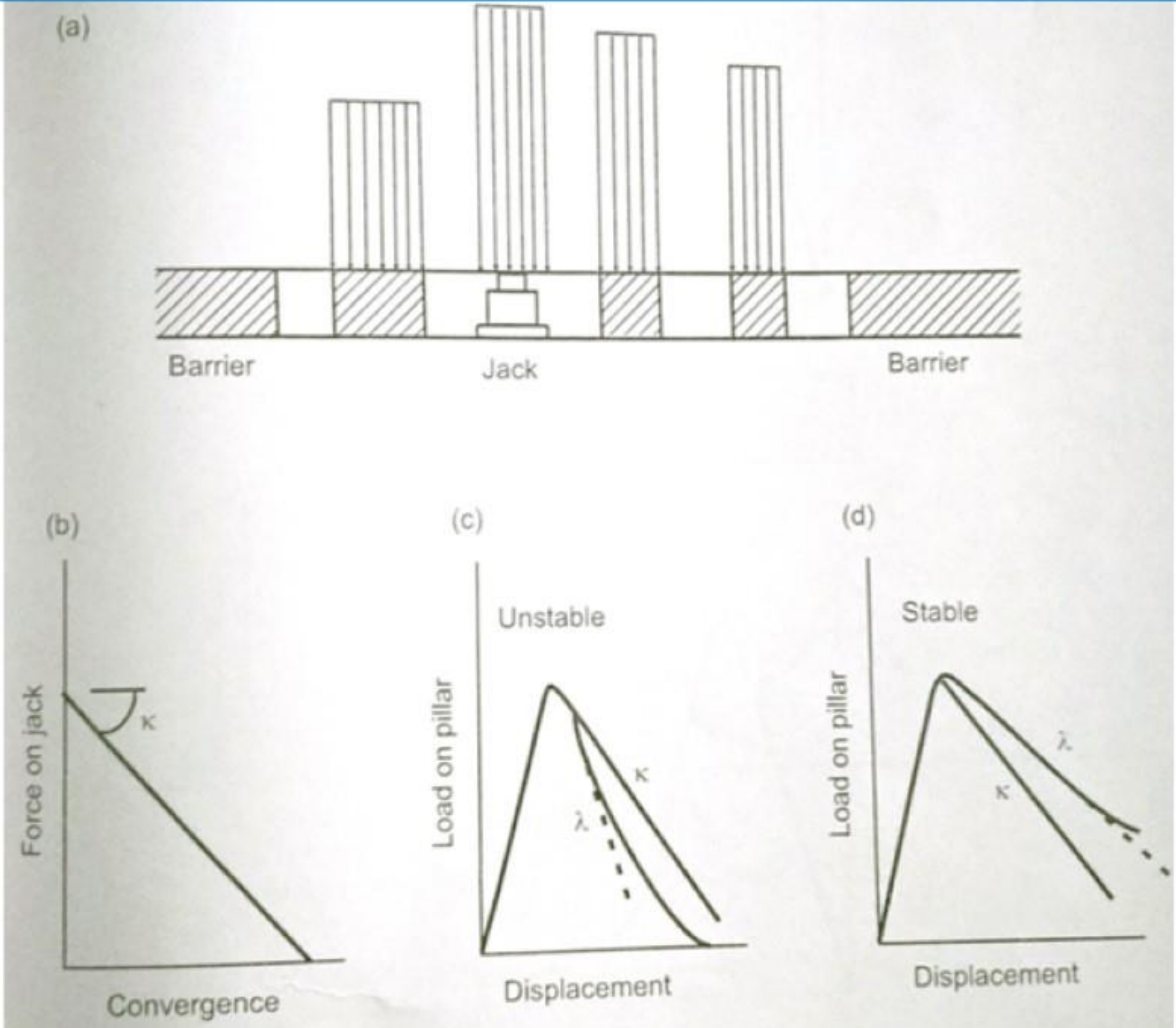


Fig. 10.9. (a)-(d) Scheme of Starfield and Fairhurst (1968) for stable and unstable pillar failure in terms of the local stiffness of the roof.

YIELD PILLAR TECHNIQUE

- ✓ If the safety factor of this pillar is less than 1.0 the failure will be stable of unstable depending on whether the local stiffness is steeper or gentler than the steepest part of the post failure stiffness of the pillar, **Figurea 10.9c and 9d** .
- ✓ Thus for stable pillar failure,

$$\kappa \geq \lambda$$

- ✓ In an array of uniform pillars surrounded by wide chain pillars, the local stiffness will have the greatest negative value next to the chain pillars and the least at the panel centre.
- ✓ Salamon (1970) has shown that pillar failure cannot be stable if a pillar undergoes tributary area load.
- ✓ This means that for the yield pillar method to succeed, panels must necessarily be narrow along one side but as long as desired along the other side.
- ✓ Also, numerical modelling will be required for estimating the load on pillars as well as the local mine stiffness.

YIELD PILLAR TECHNIQUE

Procedure for Design

- ✓ The procedure for designing the yield pillar technique for a panel of bord and pillar mining will thus consist of two steps:
  - ✓ (i) Numerical modelling for pillar load and local mine stiffness.
  - ✓ (ii) Determination of the realistic post-failure pillar characteristic, whose steepest slope should be less than the local stiffness.
- ✓ Any suitable numerical method may be used, trying out several panel widths and pillar sizes to ensure that the pillar(s) at the centre of the panel does not undergo the tributary area load and that its safety factor is somewhat less than 1.0.
- ✓ When these requirements are satisfied, the numerical model has to be run again with the central pillar(s) being removed.
- ✓ The two models will give the roof-to-floor convergence C_p, with the pillar in place, C_e with the pillar removed and σ_z the average vertical stress on the pillar or pillar load.

Procedure for Design

- ✓ The local mine stiffness is then calculated as $\kappa = \frac{\sigma_z A}{C_e - C_p}$, where A is the plan area of the pillar.
- ✓ This design method assumes that each pillar offers resistance to the roof and floor at a point and not over an area, as reflected in Equation which involves the mean vertical stress and the mean convergence values.
- ✓ This can at best be acceptable for slender pillars.
- ✓ Since pillar failure proceeds from the outer edges to the pillar centre, each element of the pillar experiences higher horizontal stress as the failure travels inwards.
- ✓ Then again, the horizontal stress falls partly due to the adjoining crushed coal, or wholly, if spalling takes place.
- ✓ Thus, in a wider pillar, a situation shown in **Figure 10.10** may arise when the pillar has a steeper post-failure curve than its replica tested in the laboratory.
- ✓ A realistic estimation of the post: failure characteristic of wider pillars (w/h > 4-6) becomes rather difficult.

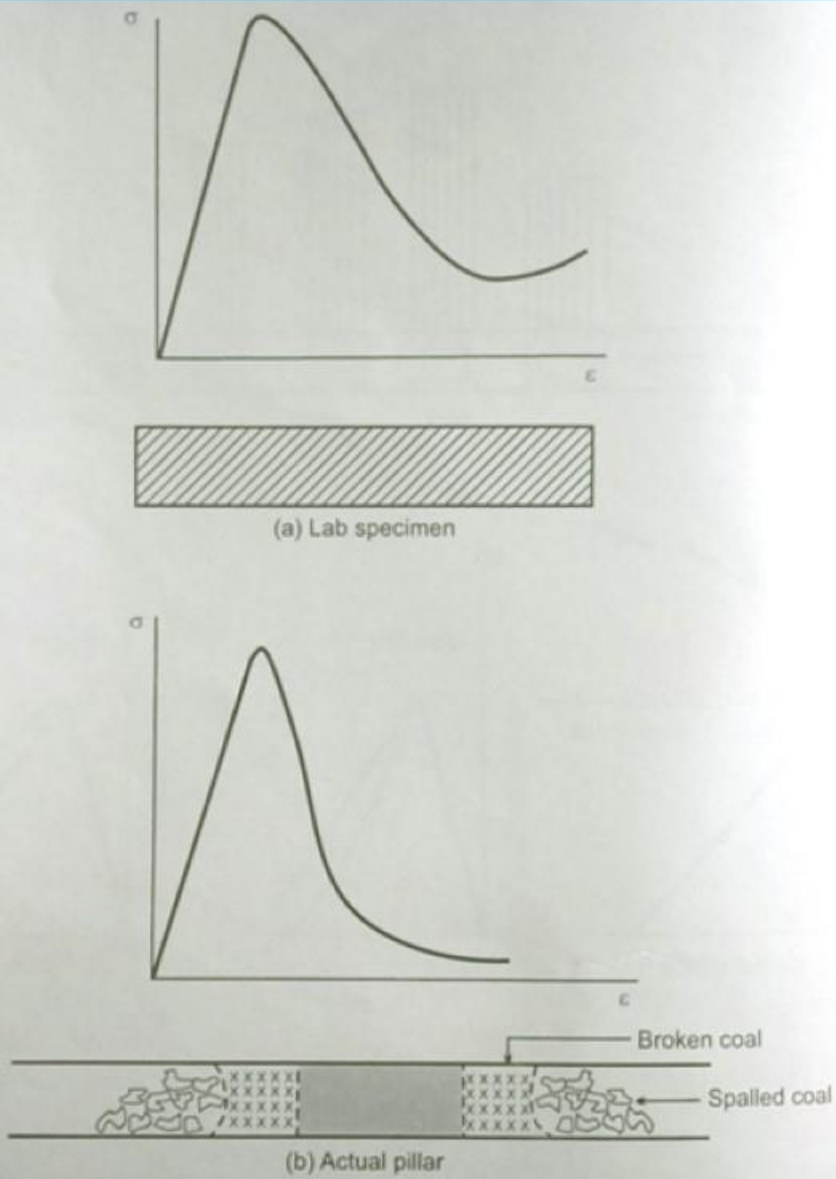


Fig. 10.10. Possible difference between the post-failure stiffness of a wide pillar and the corresponding laboratory specimen due to progressive failure in the pillar.

YIELD PILLAR TECHNIQUE

Procedure for Design

- ✓ Two procedures can be given for determining the post-failure pillar stiffness in the light of the above discussion:
- ✓ 1. Test replicas of the pillar using the same coal cither in a closed-loop servo machine in the laboratory or in situ using servo-controlled multiple jacks as done by Wagner (1974). The second test, though better, is cumbersome and expensive.
- ✓ 2. Numerical modelling of the pillar with strain softening and with realistic inputs of insitu stresses and other properties.

YIELD PILLAR TECHNIQUE

Yielding Chain Pillars

- ✓ Application of yield pillars in longwall mining has received much attention in U.S. coal mines (Carr et al., 1984; Harvey and King, 1985; Agapito et al., 1988; Serata et al., 1986; Mark, 1988; Agapito and Goodrich, 2001) and research in this area has addressed the following problems :
  - ✓ - Control of rock bursts
  - ✓ - Floor heave
  - ✓ - Stress related roof falls
- ✓ Schissler et al. (2003) have compiled 38 cases of yield pillars in longwall workings from U.S, coalfields.
- ✓ They have declared many of these cases as unsuccessful assigning the following reasons:
  - ✓ Pillar sinking in a floor with low bearing capacity.
  - ✓ Rock burst conditions
  - ✓ Weak roof yield pillars require roof strata capable of bridging.
  - ✓ Strong coal resistant to yielding.

Yielding Chain Pillars

- ✓ All these reasons fit well with the requirement that the local stiffness must be greater than the post-failure pillar stiffness for yield pillar application to be successful.
- ✓ If we consider a yield pillar row beside a wide, stable row of chain pillars (**Figure 10.11**), it can be realized that the local mine stiffness at the yield pillar location is likely to be less because of the presence of goaf on one side than if the yield pillar were in a bord and pillar panel bounded by wide chain pillars.
- ✓ Estimating the local stiffness in this situation will be quite difficult.
- ✓ Considering the Indian scenario of bord and pillar or longwall mining, chain pillar sizes are chosen generally as per the Coal Mines Regulations.
- ✓ Such chain pillars in general are likely to be unstable when between two goafs and may therefore start yielding either as the second panel extraction proceeds or upon its completion (**Figure 10.12**).
- ✓ However, since yield pillars require bridging strata to transfer some stresses to the abutments, controlled failure of such yielding chain pillars may not be possible if the conditions are burst-prone.
- ✓ An important point regarding the concept of the local mine stiffness needs to be brought out.

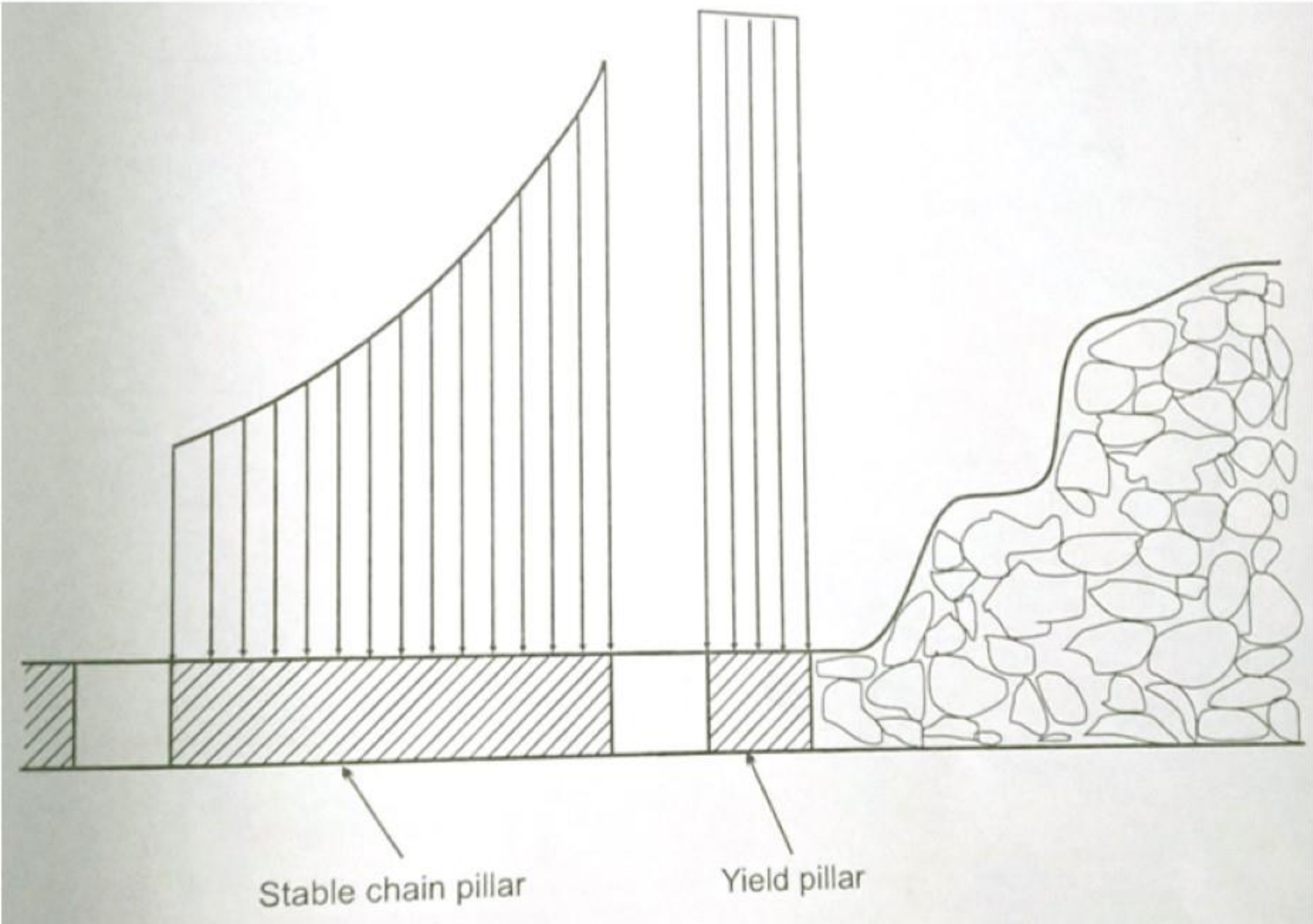


Fig. 10.11. Yield pillar beside a chain pillar in longwall.

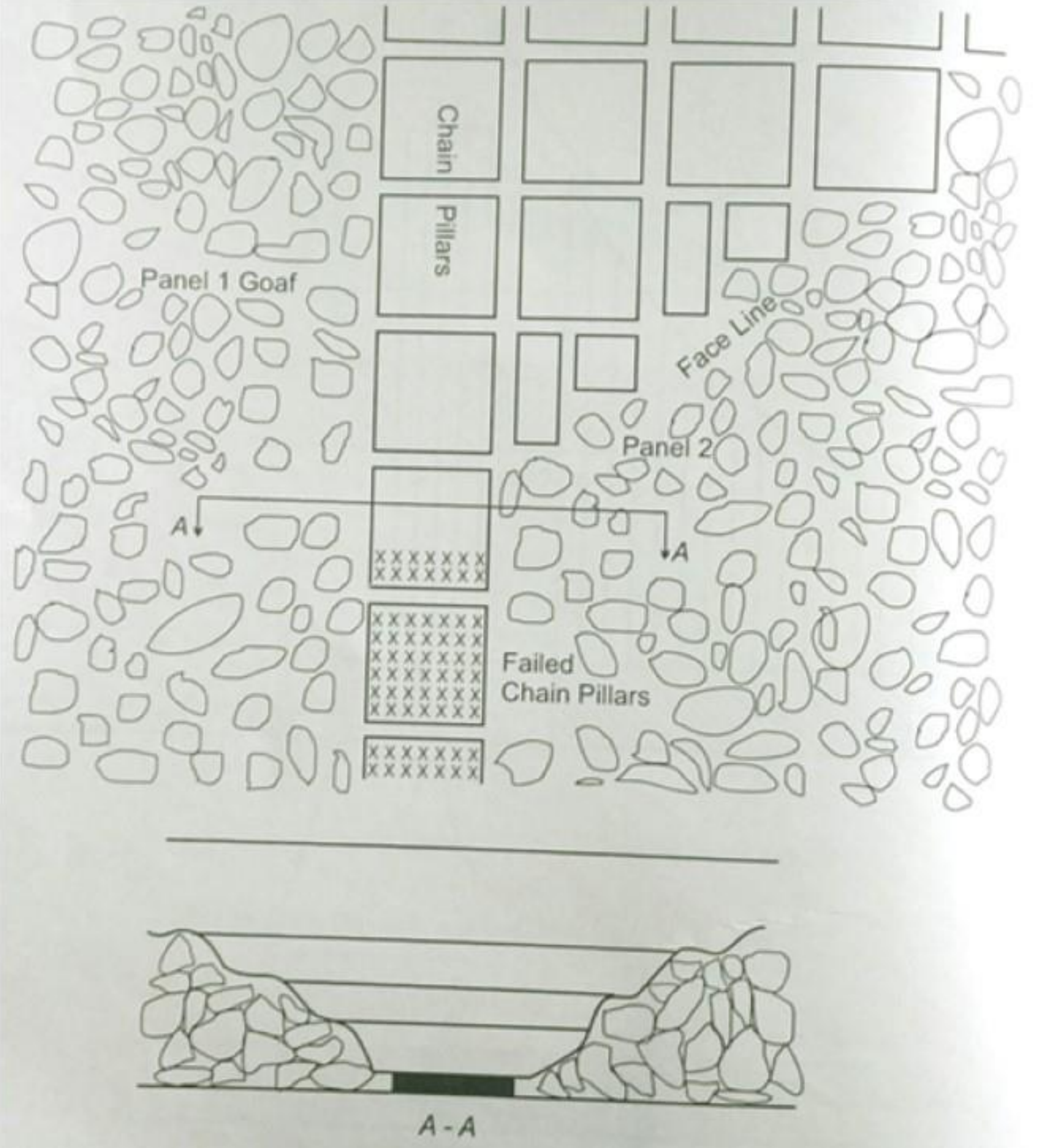


Fig. 10.12. Likely condition of chain pillars between two depillaring panels.

Yielding Chain Pillars

- ✓ While this concept is accepted and also proven for extraction under hard strata, for soft caveable strata it is redundant since such strata are not likely to cause air blasts or rock bursts.
- ✓ In fact, under well cavable strata it will often be found that

$$\kappa < \lambda$$

- ✓ As shown in **Figure 10.9c.**
- ✓ The yield pillar technique can also be useful under incompetent rock which create roof control difticultics.
- ✓ For depillaring under such conditions it will be prudent to leave judicious yield pillars in the goaf.

GOAF EDGE BLASTING

- ✓ Blasting of hard overhanging strata at the goaf edge can be one of the alternatives to avoid air blasts.
- ✓ In this technique it is by no means intended to bring down the entire overhanging strata.
- ✓ It is only meant to cause a blockage with broken rock so that air cannot escape outbye in the event of an air blast.
- ✓ It can also be used as a regular measure during extraction.
- ✓ The method is more amenable to application in those mines where long hole rock drills (Jumbo drills) are available.
- ✓ The CMRI, Dhanbad has used goaf edge blasting successfully in 4 mines where air blast problems existed (Pal Roy et al., 1996; Pal Roy et al., 2003; Sawmliana and Pal Roy, 2003; Sawmliana et al, 2004).
- ✓ All these mines have been practising the blasting gallery method of depillaring very thick seams (7-12 m).
- ✓ The method was designed by Charbonnage de France and employs Jumbo drills so that goaf edge blasting could be conveniently studied at these mines.

GOAF EDGE BLASTING

- ✓ The work at one of these mines, whose relevant details are given below, is discussed here.
  - ✓ Mine: Godavari Khani No.8 Incline
  - ✓ Owners: Singareni Collieries Companies Limited
  - ✓ Seam: No. 3 Seam
  - ✓ Seam thickness: 10.7 m (with a 2.75 m shaly band)
  - ✓ Depth of cover: 288-309 m
  - ✓ Compressive strength of roof, σ_z = 15-36 MPa
- ✓ The seam has been developed in two sections of 2.7 m each with a parting of about 5 m.
- ✓ The method of working consisted of splitting of pillars along levels and blasting the stooks so formed along with the Coal roof using a fan-shaped drilling pattern (**Figures 10.13 & 10.14**).
- ✓ The support system comprised girders placed on hydraulic props.
- ✓ The first caving area was generally about 1600 m^2 which was not large, but because of the excessive extraction height caused an air blast.

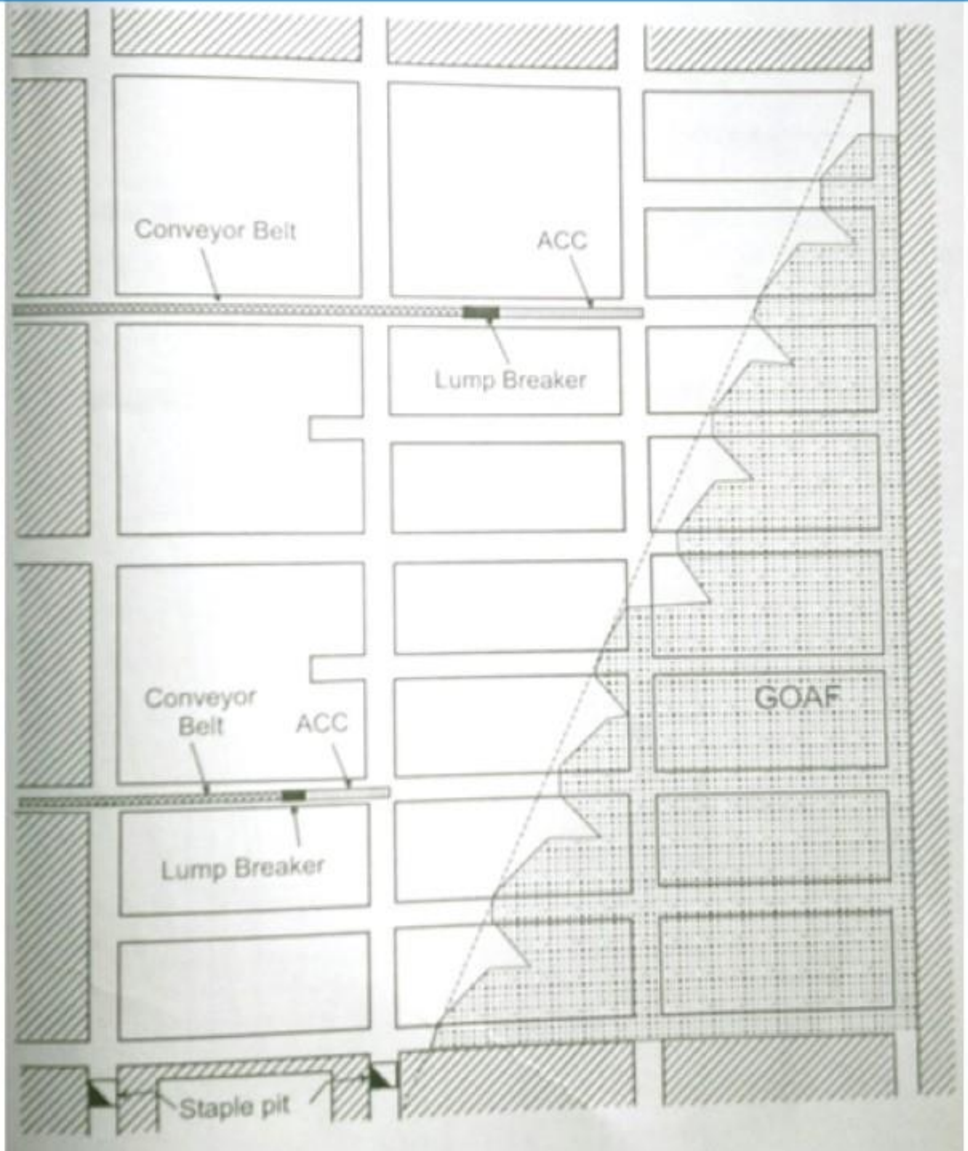


Fig. 10.13. Panel 2 for blasting gallery method of depillaring at Godavari Khani No. 8 incline (after Pal Roy *et al.*, 2003).

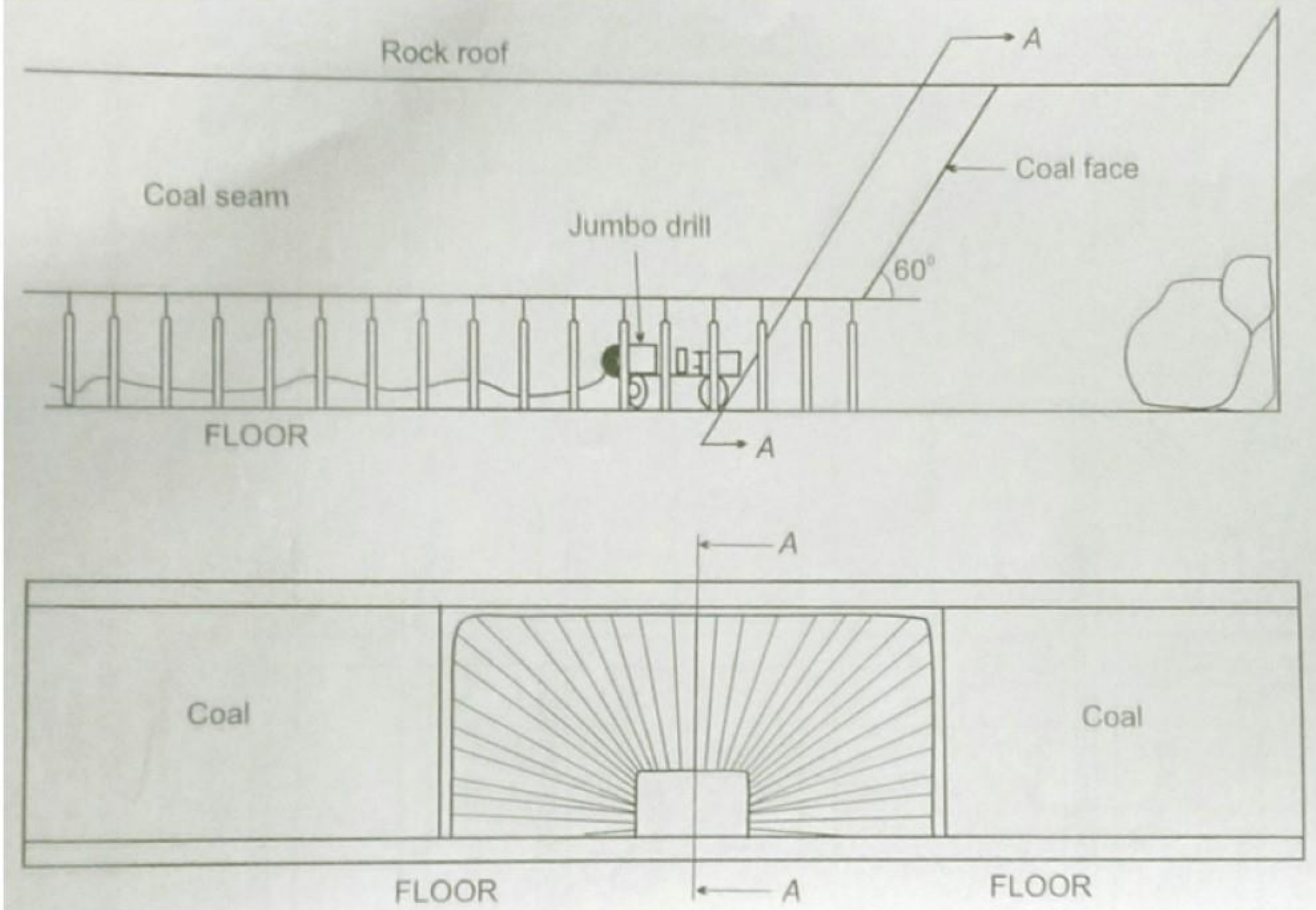


Fig. 10.14. Pattern of drilling in blasting gallery method for coal winning
(after Sawmliana and Pal Roy, 2003)

GOAF EDGE BLASTING

- ✓ After trying out some initial drilling patterns to bring down the exposed rock in the goaf, the pattern shown in **Figure 10.15** with all details was finalized.
- ✓ A length of bamboo was used to cover the coal roof and the rest of the hole was fully charged without decking.
- ✓ The fragmentation was studied by image processing of the broken rock using a software FRAGALYST developed at CMRI and was found to be satisfactory.
- ✓ The choking of all the roadways with broken rock was also acceptable.

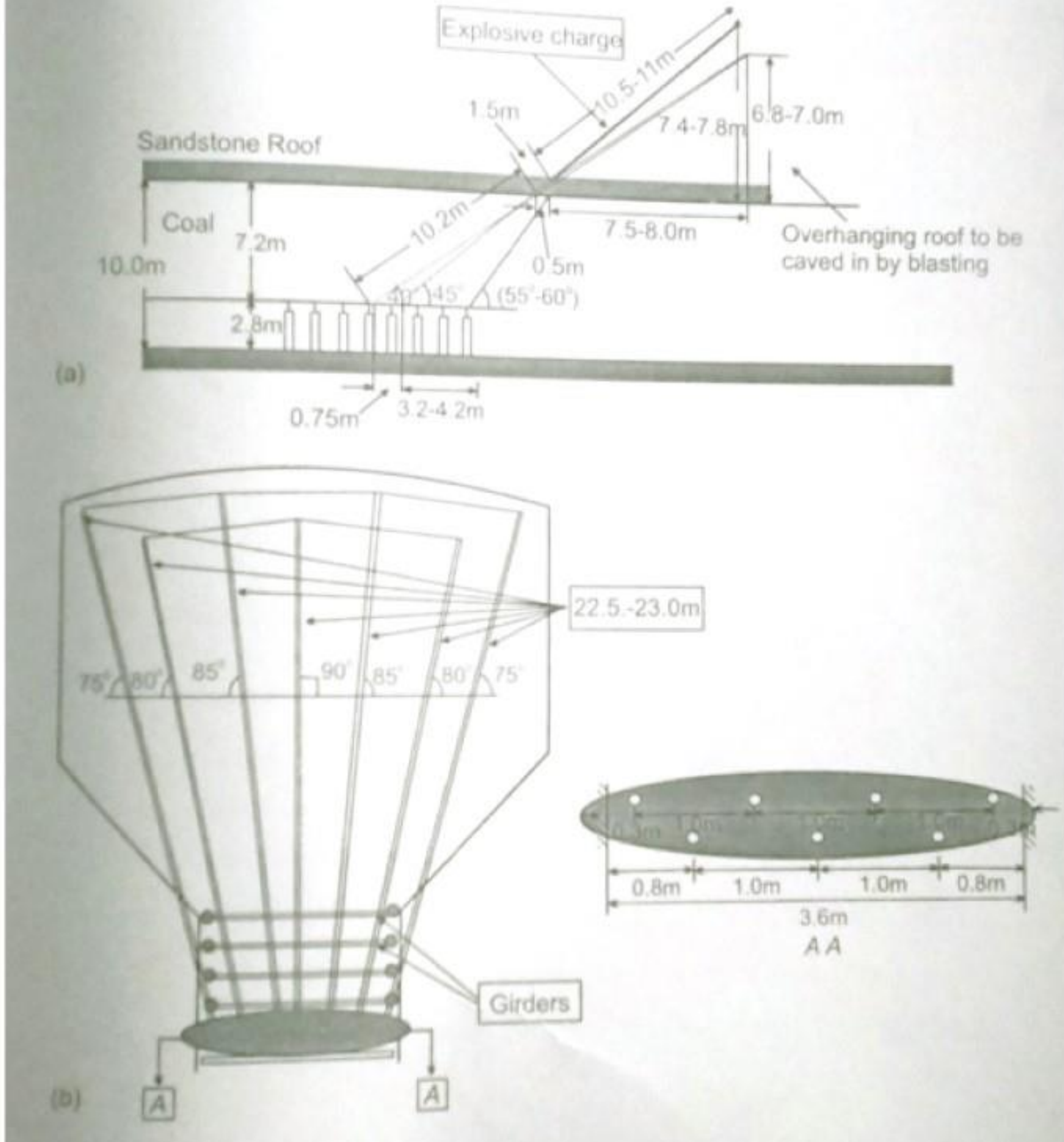


Fig. 10.15. Drilling pattern for rock caving at the goaf edge (after Pal Roy *et al.*, 2003).

GOAF EDGE SUPPORT MEASURES

- ✓ An age-old convention among mining engineers is to provide strong supports at the goaf edge to provide a “cutting edge” or breaker line with the hope that
  - ✓ caving may be better induced and
  - ✓ overriding may be avoided.
- ✓ Overriding is a term meaning encroachment by the goaf into the working area in depillaring.
- ✓ It can occur due to poor caving, inadequate or weak pillars/ribs and poor supports.
- ✓ In the Indian contest, the cutting edge is provided by the rib which protects the slicing area and by the goaf edge supports outbye of the rib.
- ✓ These supports are generally wood chocks but can be alternatively, cost permitting, hydraulic chocks.
- ✓ Rock bolting is a much cheaper and easier alternative goaf edge support and has been tried with some success in South Africa and Australia (Buddery, 1989; Vervoort, 1989; Shepherd and Chaturvedula, 1992; Singh et al., 1999).
- ✓ The rock mechanics study and the experience from goaf edge bolts can be described through the Australian case study of Blue Mountains colliery (Singh et al., 1999).

GOAF EDGE SUPPORT MEASURES

- ✓ The study panel of mechanized bord and pillar here was 230-260 m deep with a seam thickness of 1.6-2.1 m.
- ✓ The pillars were 20-35 m x 37.5 m between roadway centres with a roadway width of 6 m and a fender width of 6.5-7 m.
- ✓ The mine had a history of overriding which could encroach the working area by 5-15 m.
- ✓ The roof was a generally competent sandstone which caved at a low angle of 25°, this angle with the horizontal being the overall angle of hanging strata after caving.
- ✓ **Figure 10.16** shows a layout of goaf edge supports consisting of a row of 1.5 m resin grouted instrumented rock bolts, which had 18 strain gauges in two slots, followed by a row of point anchor bolts and breaker props.
- ✓ A roof extensometer and a convergence indicator along with 4 plastic "shear" tubes in roof holes were also installed.
- ✓ During caving it was observed that overriding was arrested successfully by the rock bolts by effectively containing the shear planes cutting across the bolts, as is apparent from the bending and axial strain plots of **Figures 10.17 & 10.18** obtained from an instrumented bolt.
- ✓ It was thus seen that rock bolts could control overriding quite satisfactorily.
- ✓ Since it is impossible to predict the exact position of the face line at which caving will occur it will be necessary to cover the entire depillaring panel by bolting to avoid overriding.

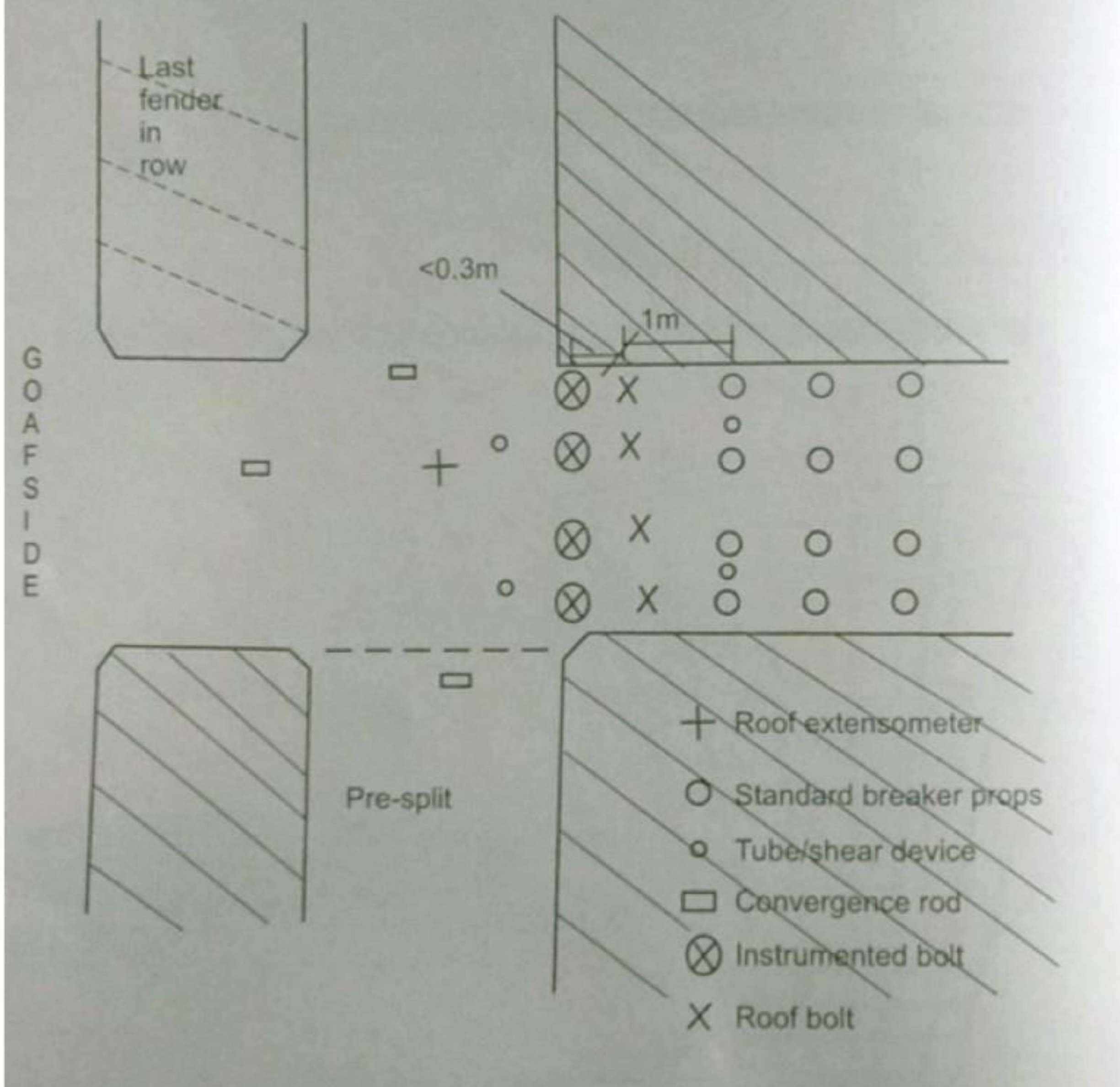


Fig. 10.16. Layout of bolts, props and instruments at the goaf edge (after Singh *et al.*, 1999).

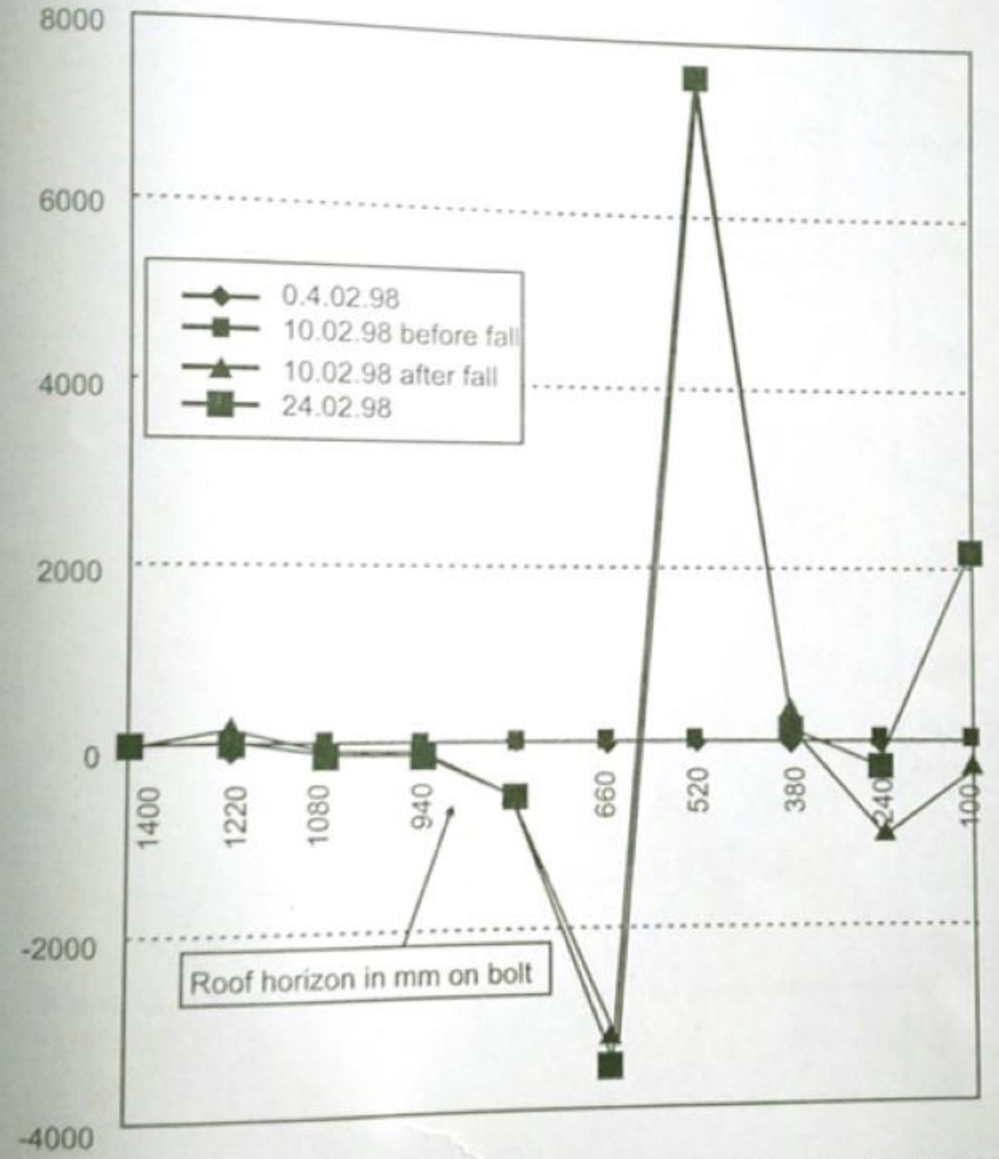


Fig. 10.17. Bending strain along instrumented rock bolt in the roof (after Singh *et al.*, 1999).

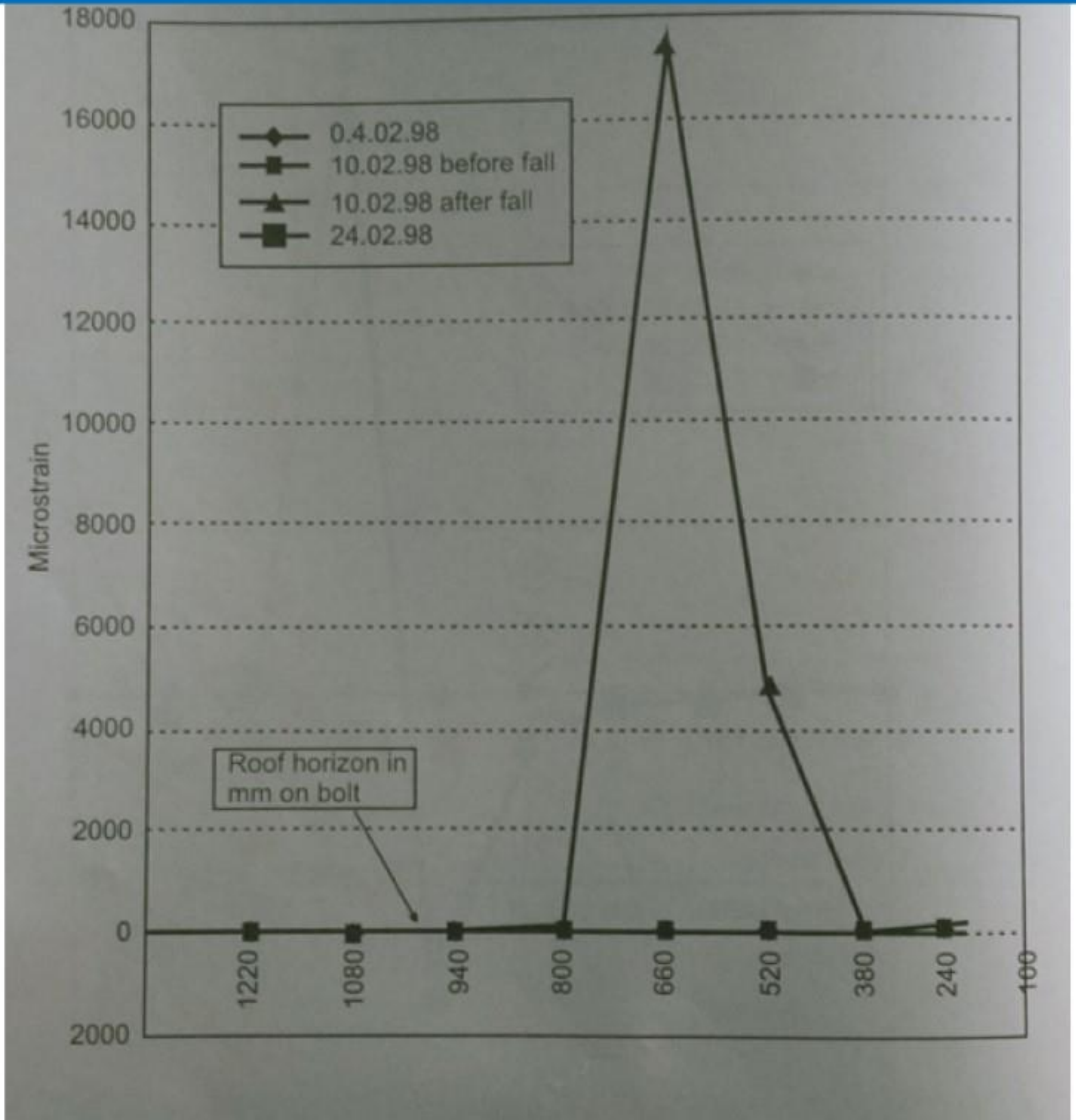


Fig. 10.18. Axial strain along instrumented rock bolt in the roof (after Singh *et al.*, 1999).

ROCK BURSTS

- ✓ As mentioned in the beginning of this Chapter, rock bursts in Indian coal mines are confined to Raniganj coalfield.
- ✓ However, with finding of new deep seams and as mining goes deeper, this problem can assume greater importance.
- ✓ The following 5 conditions are necessary for rock bursts to occur (Crouch and Fairburst, 1974).
  - ✓ Depth of cover generally exceeding 300 m.
  - ✓ Strong, stiff roof strata which are not easily cavable.
  - ✓ Structurally strong coal, not necessarily with a high compressive strength.
  - ✓ Strong floor which cannot heave easily.
  - ✓ Improper mining method producing high stresses.
- ✓ Besides these conditions, proximity of major geological features may contribute in inducing bursts.
- ✓ Only factor 5 above is controllable.

ROCK BURSTS

- ✓ The frequency of bursts in American coal mines was categorized based on mining method and place of occurrence as given in Table (Haramy et al., 1985).
- ✓ Table: Frequency of burst occurrence.

Mining Method	Frequency %	Place of occurrence	Frequency %
Bord and pillar	74.7	Mine entries	76.3
Longwall	22.1	Working face	10.5
Others	3.2	Longwall face	9.4
		Intersections	3.0
		Others	0.8

- ✓ According to an analysis (Singh and Singh, 1988) carried out for 94 incidents of burst in Indian coal mines between 1944 and 1964, depillaring accounted for 85% of the bursts (58.4% on faces and 26.6% in roadways), whereas a single longwall panel at Parbelia colliery alone contributed 9.5% of the bursts.

Burst proneness of coal and detection of bursts

- ✓ Several indices based on the mechanical properties of coal have been proposed (Kidybinski, 1981; Krzyszton, 1992; Benyi et al., 1986; Singh, 1987) to define burst proneness.
- ✓ In the first of these references are given several indices based on polish research carried out at the Central Mining Institute (CMI), Katowice.
- ✓ Of all indices, W_{ET} has been widely accepted because of simplicity and logic of definition.
- ✓ The total energy imparted to the specimen is composed of two components: one is elastic strain energy E_e, which is released immediately upon release of applied stress and the other is permanent strain energy E_p, which is irrecoverable upon release of applied loading (**Figure 10.19**).
- ✓ W_{ET} is simply defined as (Szekowka et. al., 1973) as $W_{ET} = \frac{E_e}{E_p}$
- ✓ Using this index, liability to bursts is classified as follows:

Degree of bump liability	W_{ET}
Severely liable	> 5
Slightly liable	2-5
Not liable	< 2

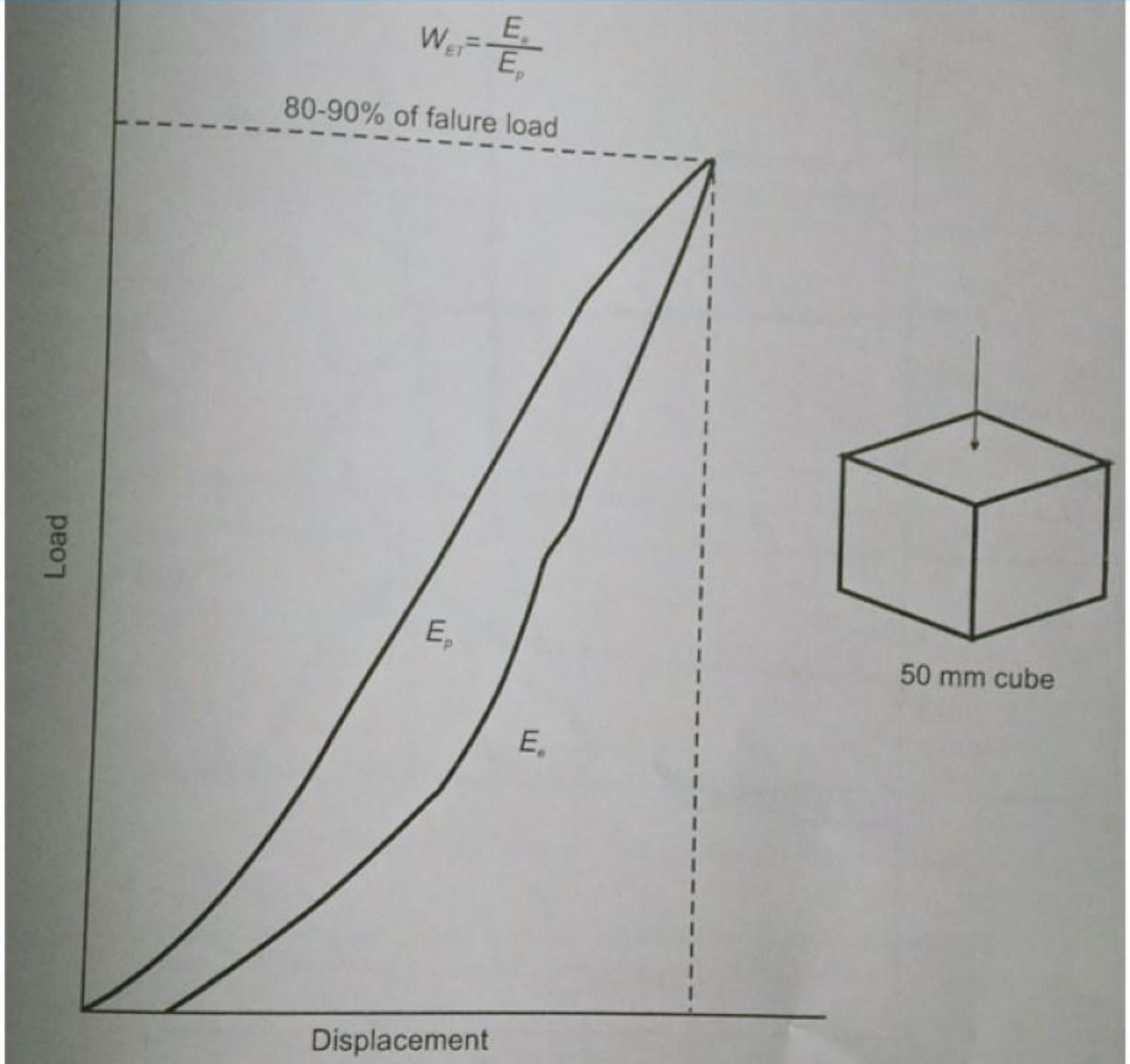


Fig. 10.19. Definition and procedure for energy index of coal (after Szekowka *et al.*, 1973).

Burst proneness of coal and detection of bursts

- ✓ As an alternative to the above Polish methods, susceptibility to bumps can also be estimated using the following two proposed methods.
- ✓ According to the Czech method (Zamarski and Franek, 1977) a seam is bump prone if the following two conditions are simultaneously satisfied:
  - ✓ 1. Elastic strain limit of coal ≥ 70%
  - ✓ 2. K ≥ 3.0
- ✓ where $K = K_1 + K_2 + K_3$; $K_1 = \frac{Depth\ of\ cover\ (m)}{300}$; $K_2 = \frac{Thickness\ (m)\ of\ roof\ strata\ of\ strength > 700 kg/cm^2}{10}$;

  $K_3 = \frac{Average\ strength\ of\ immediate\ strata\ (kg/cm^2)}{700}$
- ✓ The thickness of the immediate strata is taken five times the seam thickness.
- ✓ The other method, proposed by Brauner of Germany (1988), consists of keeping a cubic coal specimen under total confinement in steel, giving a fixed load on top and drilling a small hole at its side.

ROCK BURSTS

Burst proneness of coal and detection of bursts

- ✓ The resulting stress relaxation with drilling is plotted on a closed-loop servo control machine.
- ✓ If the relaxation curve shows steps, the coal is bump prone.
- ✓ Each of the indices quoted above leads to classifications or criteria for the bursting liability of coal.
- ✓ However, for complete predictions of rock burst or bump hazard for a specific location in a mine, data on **insitu stress** concentrations must also be collected.
- ✓ Since direct stress measurements in coal are not recommendable as a routine procedure in mines, a number of indirect detecting methods have been tried which include: deformation measurements method, drilling yield method and microseismic method.

YIELD PILLAR TECHNIQUE

Yielding Chain Pillars: Drilling yield method

- ✓ The drilling yield method is widely used throughout the world to locate zones of abnormal or high stress concentration (Kidybinski, 1981; Neyman ef al., 1972; Noack et al., 1983; Brauner, 1989).
- ✓ This method, being used in the USSR since the 1950's, was modified and adapted to local geological conditions in the early 1960's in a few of the European coalfields.
- ✓ The method was later introduced to U.S. coal mines and was favorably accepted (Haramy and McDonnel, 1988).
- ✓ This method involves drilling holes into the seam and recording the volume of drill cuttings obtained from each metre of a given hole depth.
- ✓ A certain volume of cuttings can be expected from a drill hole that has a known diameter and length.
- ✓ If the actual volume of cuttings generated exceeds the volume of the hole by a significant amount, the zone around that particular hole is determined to be highly stressed.
- ✓ The critical values for drilling yields applied in Germany (Noack et al., 1983) are 90 l/m for 140 mm diameter boreholes; 50 l/m for 95 mm diameter holes; 8 l/m for 50 mm boreholes and 6 l/m for 42 mm boreholes.

YIELD PILLAR TECHNIQUE

Yielding Chain Pillars: Drilling yield method

- ✓ Above these values, the stress concentration is dangerous and a critical state of burst hazard is assumed.
- ✓ It is to be noted that such high yields of dust must be accompanied by drill jamming, grating etc.
- ✓ A typical test result is shown in **Figure 10.20**.
- ✓ Studies carried out in Germany and Poland provided a basis for stating a relationship between burst potential determined from drilling yield and extracted seam thickness.
- ✓ If a highly stressed zone is detected at a distance of more than 3.4 times the seam thickness from the rib side, there is believed to be no danger but when the zone is less than 1.5 times the seam thickness away an extremely hazardous state exists and men should be withdrawn.

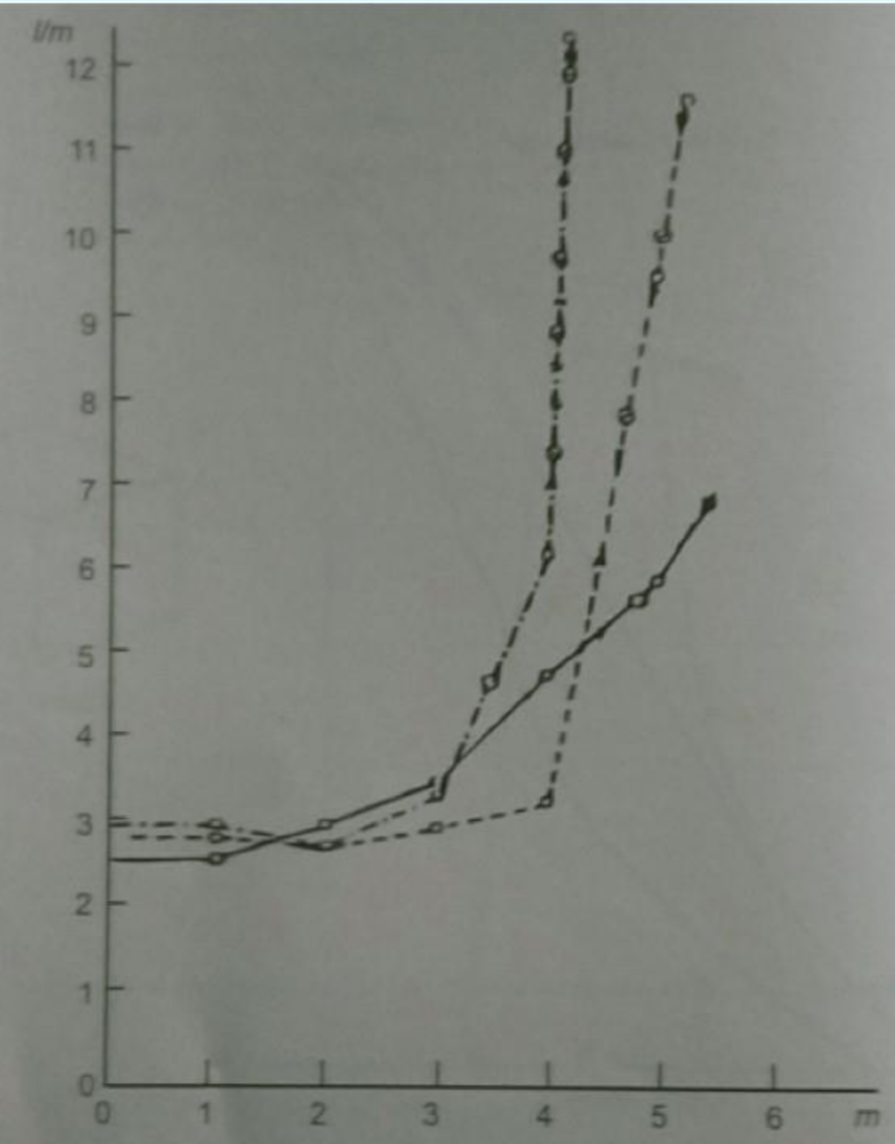


Fig. 10.20. Drilling yield test result in a burst-prone situation (after Neyman *et al.*, 1972).

YIELD PILLAR TECHNIQUE

Yielding Chain Pillars: Microseismic method

- ✓ The microseismic method, also known as the acoustic emissions (AE) method, was developed by the U.S. Bureau of Mines in the late 1930's (Obert and Duvall, 1945; 1967).
- ✓ It is a geophysical approach that detects subaudible rock noise (microseisms) associated with yielding or fracturing of the rock.
- ✓ Geophones translate the rock noises from acoustic waves, which may be undetectable by the unaided ear, to electrical analogue signals.
- ✓ Laboratory studies of rock indicate increased noise rate with increased stress, the rate of increase becoming pronounced as the ultimate stress is approached.
- ✓ A pattern of increased activity followed by a quiet period immediately prior to a bump has been noted by several investigators, but this behaviour has hot been consistently reproduced in the laboratory (McCabe, 1978; Chugh and Heidingcr, 1978).
- ✓ While some bumps are preceded by rapid increases in the microscismic noise rate, others show no rate increase, and sometimes, while rapid rate increases are measured, no bumps occur (Jackson, 1984; Kneislcy, 89).

YIELD PILLAR TECHNIQUE

Yielding Chain Pillars: Microseismic method

- ✓ Large scale microseismic research in Polish coal mines began in 1963; these investigations have contributed to reducing bumps in the Upper Silesian Coalfield from over 300 per year to only about 20 per year (Trombik and Zuberek, 1975; 1978).
- ✓ The Polish studies indicate a period of increased microseismic activity followed by a period of decreased activity prior to coal bumps, and that bump, occurred on the background of this so-called microseismic calm.
- ✓ In the United States Leighton (1984), from an investigation of bord and pillar mining, monitored dramatic increases in microseismic activity as mining approached a fault.
- ✓ From this study Leighton could establish a better mine layout and extraction to minimize bumping.
- ✓ Microscismic studies conducted in the Olga No.2 Mine near Caretta (USA), indicated an increase up to one order of magnitude in activity prior to a massive bump that damaged about 100 pillars (Condon and Munsop, 1987).

YIELD PILLAR TECHNIQUE

Yielding Chain Pillars: Rock deformation measurement

- ✓ Rock deformation measurements such as convergence or convergence rate and pillar deformation have been successfully used to detect burst-prone situations underground at the face or in the entries,
- ✓ In German coal mines (Brauner, 1989), in order to identify a burst prone situation, the ratio of rib side extrusion and roof-to-floor convergence has been found to be a useful index.
- ✓ The case study of Chinakuri No. 1 mine gives the results of vertical pillar deformation measurements which clearly indicate the onset of a burst.
- ✓ The US Bureau of Mines carried out stress measurement in chain pillars left between two longwall panels extracted one after the other in a burst-prone seam (Iannachione, 1990).
- ✓ Convergence was also measured in adjacent entries (gateroads).
- ✓ **Figures 10.21 & 10.22** indicate that such measurements can be useful in detecting the start of bursts.
- ✓ Similar results of measurements will doubtless be obtained in a mine which is not burst-prone.
- ✓ So, the obvious requirement is that the seam conditions must be burst-prone for such underground observations to be meaningful for prediction or detection of bursts.

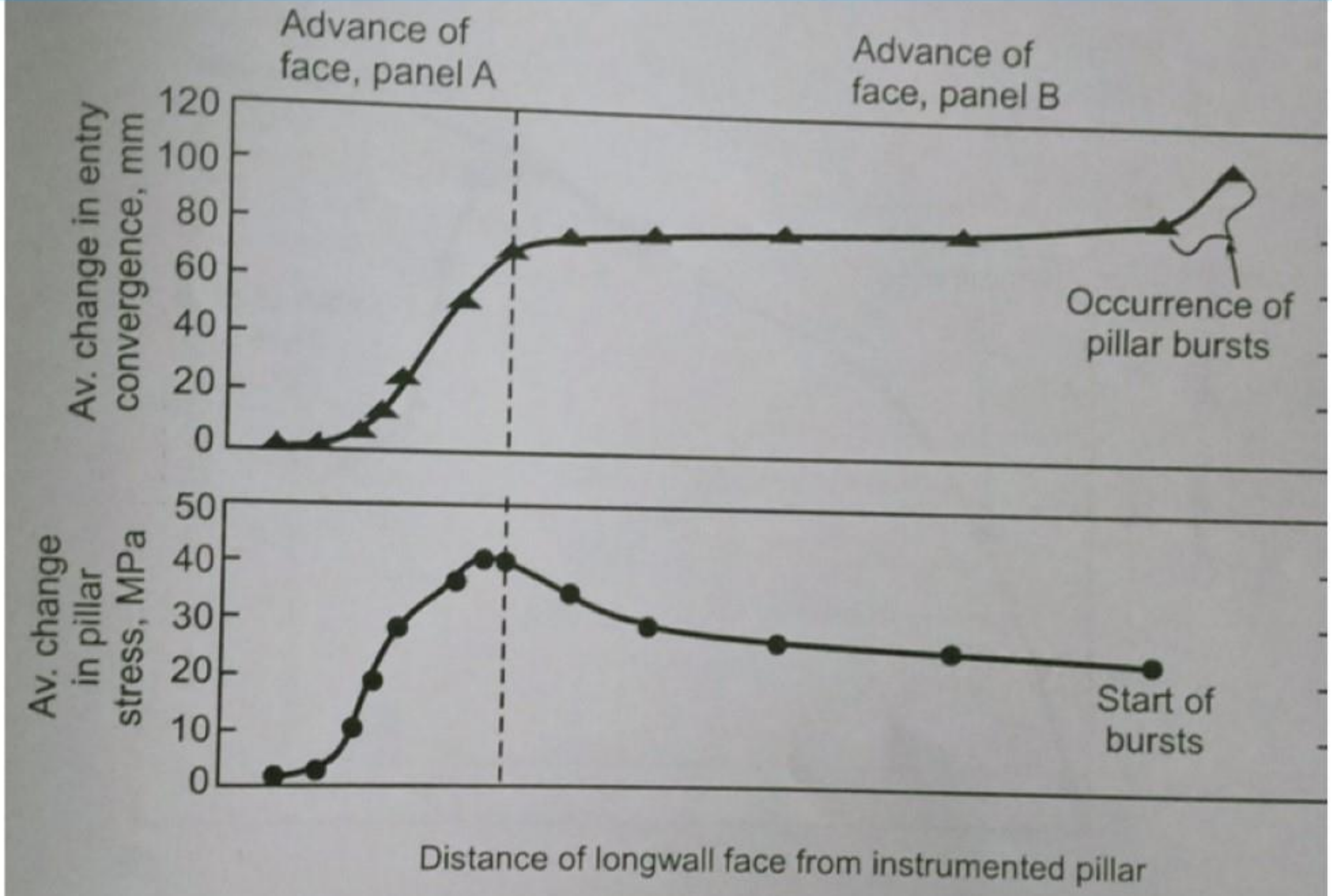


Fig. 10.21. Average change in pillar stress and roof-to-floor convergence at various face positions of longwall panels A and B (after Iannacchione, 1990).

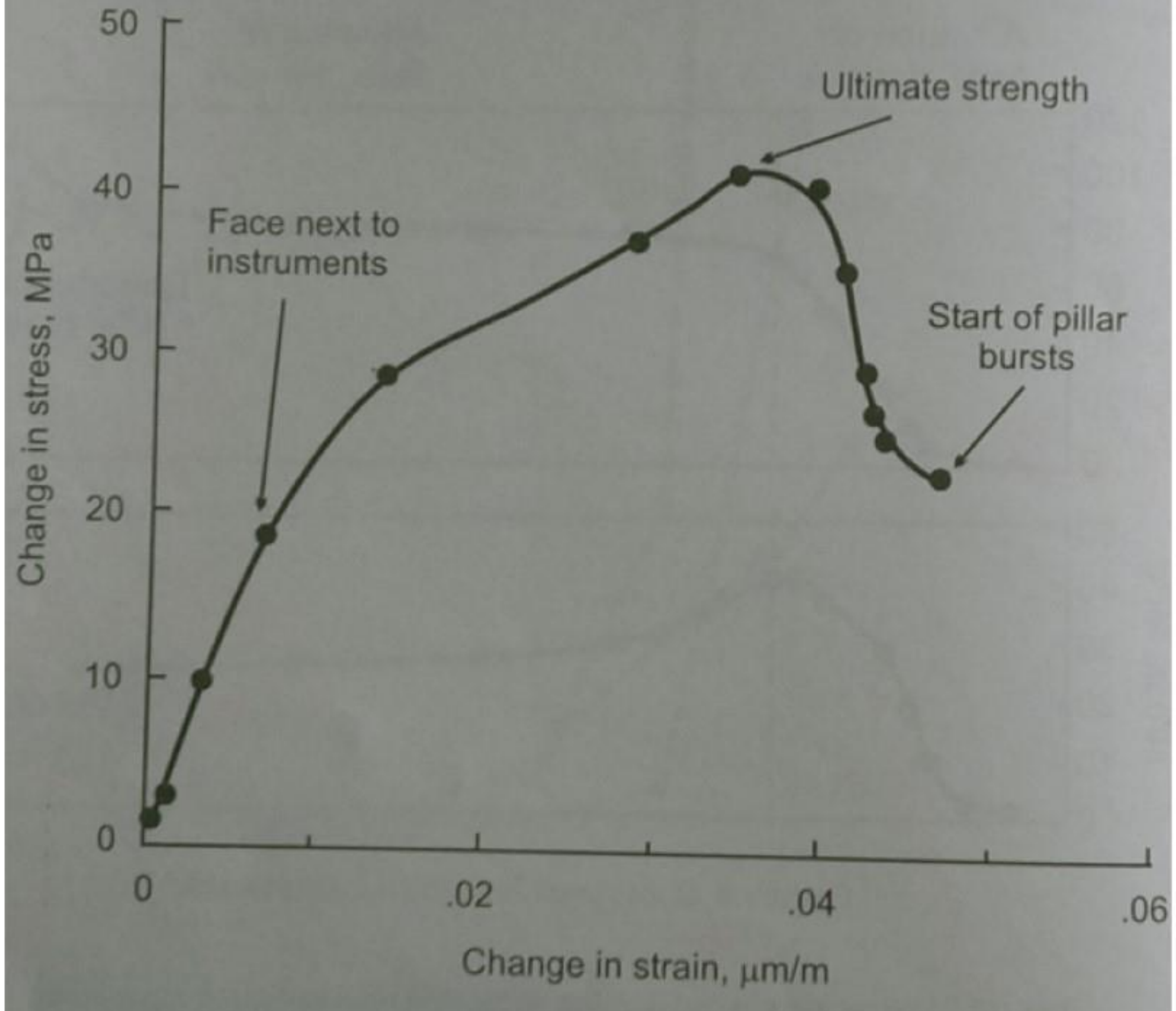


Fig. 10.22. *In situ* stress-strain curves for the Pocahontas no.3 Coalbed as the pillar was loaded to failure (after Iannacchione, 1990).

Combating and control of bursts

- ✓ The basic concept of destressing or transferring high-stress concentrations from one portion of a mine structure to another is not new for combatting bursts.
- ✓ Fracturing or softening coal or rock to control stress build up has been practised in various mines and has proved helpful (Neyman er al., 1972).
- ✓ In coal mines, destressing the active working face is the most logical method to prevent bumps.
- ✓ Coal and in some instances roof and floor rock, is intentionally fractured and made to fail.
- ✓ As a result no high-stress build up can occur in the fractured zone, and the load is transferred to another part of the mine structures.
- ✓ Three major destressing methods used in coal mines are discussed below.

YIELD PILLAR TECHNIQUE

Combating and control of bursts: Volley Firing

- ✓ Destressing by volley firing has successfully reduced the number of coal bumps or bursts in underground mines (Haramy et al., 1985).
- ✓ In this method peak stress locations are first determined from drilling data.
- ✓ For relieving stress the high stress locations of a longwall face are then drilled and blasted to destroy the integrity of the coal.
- ✓ The face is then advanced through blasted coal.
- ✓ The blast holes are loaded with 1.4 kg of permissible explosives, stemmed and detonated.
- ✓ The drill pattern consists of a series of 50 mm holes, 4-4.5 m deep, drilled on approximately 1.2 m centres.
- ✓ The hole depth depends on the required daily advance of the face and on the location and magnitude of the stress abutment ahead of the face.
- ✓ The comers of the advancing longwall face require a radial drilling pattern, using combinations of two or three holes angled at 10° and 45° to relieve high stress in both the face and rib areas

Combating and control of bursts: Hydraulic fracturing

- ✓ Hydraulic fracturing requires drilling holes about 30 m deep into the panel rib side and injecting high pressure water into the seam.
- ✓ This process fractures the coal and relieves accumulated stress.
- ✓ Though the method is time consuming, and is not recommended for use on the face, it has been found to be most effective in the roof and coal seam ahead of the longwall face (Haramy and McDonnel, 1988).
- ✓ Fracturing a very large and highly stressed area causes loads to be redistributed and may create burst conditions in the mine.

Auger drilling

- ✓ Drilling large diameter holes, 150-300 mm, using augers has been in practice in Europe and the US for stress relief in burst-prone longwall faces (Talman and Shrooder, 1958; Haramy et al., 1985; Vinokur and Nevolin, 1972; Brauner, 1988). The holes are placed 4-6 m apart and the length traverses the estimated high stress zone. Such drilling has to be done by remote operation since a burst may occur by the drilling itself.

Design of mine layouts

- ✓ In South Africa a parameter called Energy Release Rate (ERR) was defined to arrive at a mine layout design which would minimize rock bursts in tabular mineral deposits (Cook et al., 1966).
- ✓ ERR is the energy released by mining a unit area of the orebody or scam.
- ✓ When ore is extracted it becomes free of stresses, the exact equivalent of which gets transmitted to the surrounding rock, which thus undergoes an increase in stresses.
- ✓ Salamon(1984) has shown that $ERR \propto \frac{\sigma_1^2(1-\nu^2)h}{2E}$ where σ_1= major principal stress, E, ν= elastic constants, h =extraction thickness.
- ✓ It has been shown from a number of case studies (Ortlepp, 1983) that the number of rock bursts is non-linearly proportional to ERR (**Figure 10.23**).
- ✓ Planning of a mine layout thus consists of minimizing ERR by adopting the following strategies:
- ✓ Reduction in panel width
- ✓ Reduction in extraction thickness
- ✓ Stowing of voids.

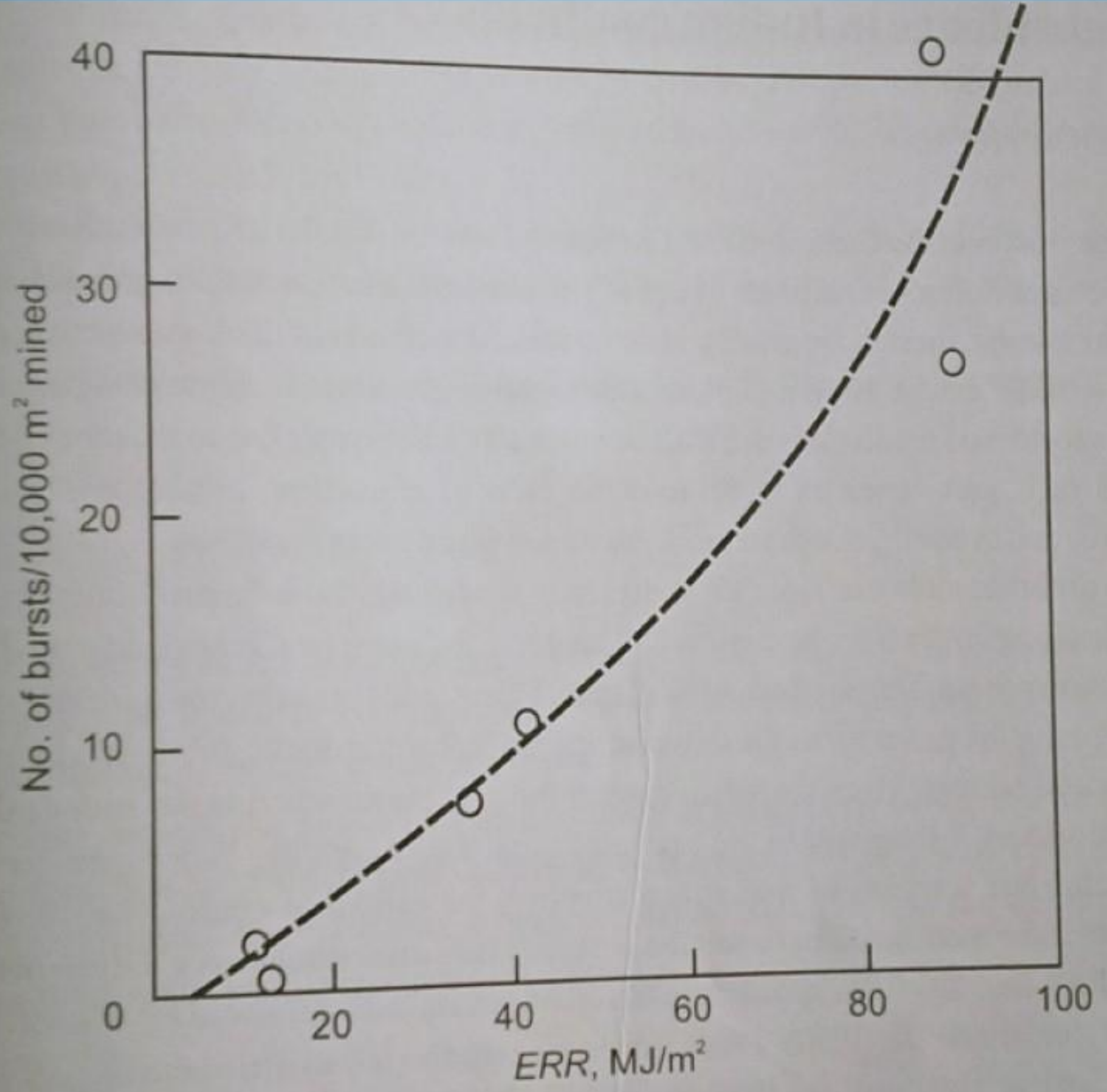


Fig. 10.23. Dependence of rock burst incidence on the Energy Release Rate (after Ortlepp, 1983).

YIELD PILLAR TECHNIQUE

Design of mine layouts

- ✓ Another method gaining increasing popularity consists of leaving carefully designed yield pillars (Campoli et al., 1987; Haramy and Kneisley, 1990).
- ✓ From Figure it may be realized that the cut-off value of ERR is about 3-4 MJ/m^2 as measured from the figure approximately.
- ✓ Although the author of this publication has not himself suggested any such cutoff value, it is reasonable to suppose that rock bursts should not take place below a certain value of ERR.
- ✓ The cut-off value will however depend on the method of extraction and local geology and also, importantly, it will be much less in the case of coal mines because coal is a much weaker material than hard rocks in metal mines.
- ✓ It may be realized that a cut-off value for coal mines is impossible to suggest without sufficient case studies.
- ✓ ERR therefore has to be accepted as a relative index which compares one design with another at the same mine.

YIELD PILLAR TECHNIQUE

Burst-proneness of Indian seams

- ✓ After a large scale testing program at CMRI involving several coal properties, the energy index W_{ET} was found to be more appropriate that other properties.
- ✓ Only four seams/mines were found to be burst-prone (Table).
- ✓ Table: Burst-prone seams in Raniganj coalfield.

Seam (Mine)	Energy index
Dishergarh (Chinakuri No.1)	4.02
Hatnal (Seetalpur)	3.46
Koithee (Bhanora)	3.17
Sanctoria (Parbelia)	2.96
Poniati (Bhanora)	1.44

CASE STUDY

ROCK BURST IN DISHERGARH SEAM OF CHINAKURI NO.1 MINE

- ✓ Chinakuri No.1 mine of Eastern Coalfields Limited is located in Raniganj Coalfield in West Bengal and has the dubious distinction of being the deepest coal mine in the country.
- ✓ Dishergarh seam which is being worked here has had a long history of seismic activity, the earliest rock burst incidents being reported in the 1920's (Ghose, 1988).
- ✓ As has been observed elsewhere in the world, seismic events at Chinakuri have far outnumbered rock burst incidents which, unfortunately, had not been monitored or analysed scientifically in the past.
- ✓ This case was perhaps the first comprehensive study of a coal mine rock burst in India (Sheorey et al., 1997).

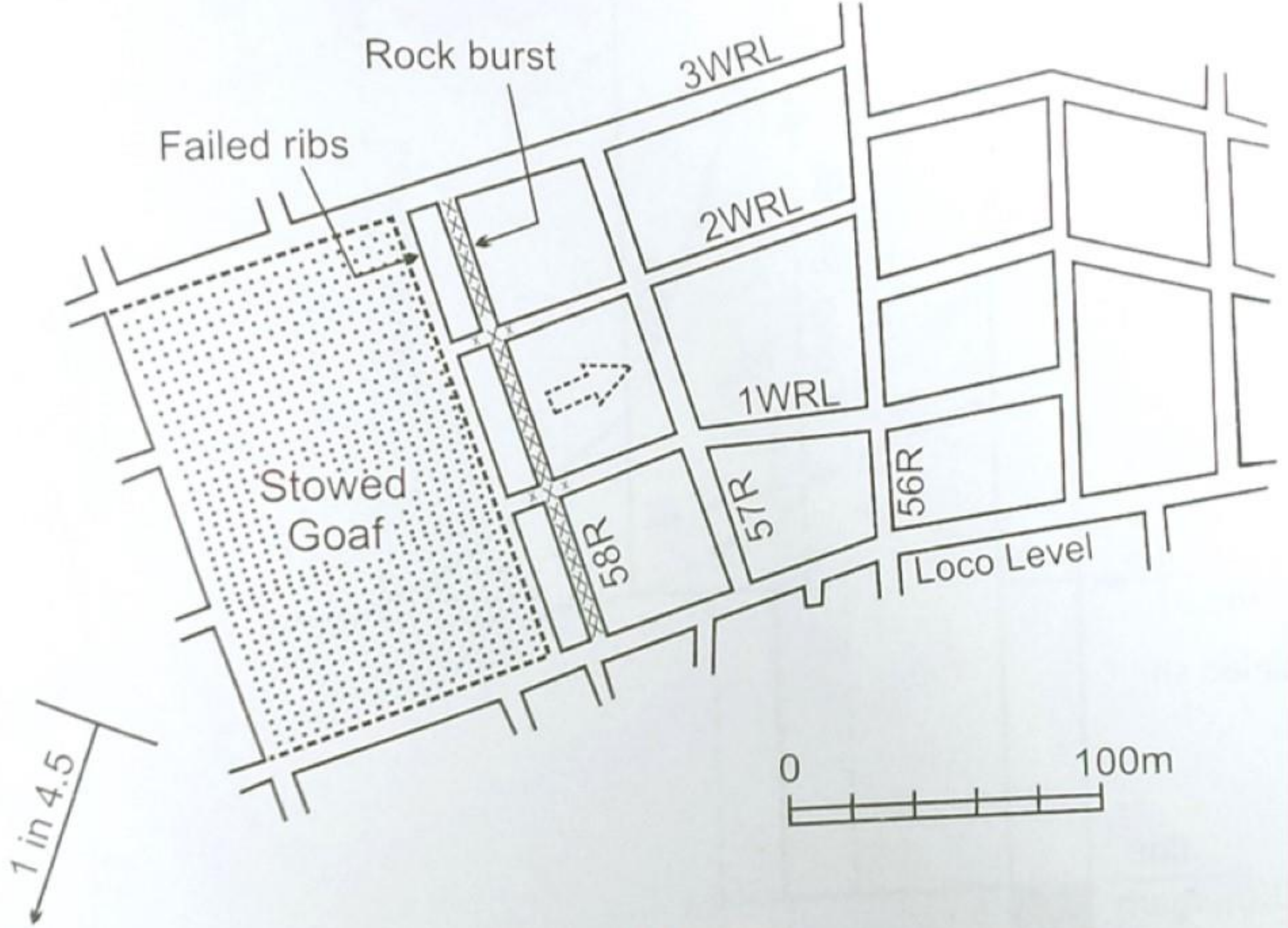


Fig. 13.12. Part plan of panel W-13, Dishergarh seam, Chinakuri no.1.

ROCK BURST IN DISHERGARH SEAM OF CHINAKURI NO.1 MINE

The site and the incident

- ✓ The extraction panel, in which the rock burst took place (Fig.13.12) on 10-10-1992 leading to 3 fatalities, had the following details:

Seam	Dishergarh
Panel	W-13
Average cover	600 m
Seam thickness	3.35 m
Extraction height	2.0 m
Gradient	1 in 4.5
Pillar size roadway centres	between 45 m (minimum)
Roadway width	3.6 – 4 m

CASE STUDY

ROCK BURST IN DISHERGARH SEAM OF CHINAKURI NO.1 MINE

The site and the incident

- ✓ The method of working was straight line depillaring (modified or "Barry" longwall) with stowing using 40 t TCR friction props and link bars as supports and a chain conveyor at the face.
- ✓ The face supports were on a 2.4 m x 1.2 m pattern and roadways ahead of the face were supported up to 30 m outbye by rope trusses at 1.2 m interval and wood chocks at 2.4 m interval.
- ✓ Coal winning was done by drilling ang blasting, each cut being 0.6 m.
- ✓ At the end of extraction of the first row of pillars, 7 m wide ribs (as per permission) were left, the chain conveyor and supports were shifted to the outbye advance roadway (59 R) and the second row extraction was started.
- ✓ When the average rib width was 8.2 m preparations for the new face were going on in 58 R and a rock burst occurred in this roadway.

CASE STUDY

ROCK BURST IN DISHERGARH SEAM OF CHINAKURI NO.1 MINE

The site and the incident: Visual Observations

- ✓ Underground inspection and discussions with the colliery management revealed the following:
- ✓ (a) The rock burst was heard as a massive booming crash quite some distance away.
- ✓ (b) About 1.5 - 2 m of roof consisting of coal and shaly sandstone had collapsed in 58R dislodging supports. Some falls had also occurred in adjoining levels.
- ✓ (c) Severe spalling, squeezing and extrusion of ribs had taken place, indicating rib crushing.
- ✓ (d) Warm water was dripping at the face, which was otherwise always dry.
- ✓ These symptoms defined this as a rock burst or, more specifically, an event comprising roof bus and coal bump.

CASE STUDY

ROCK BURST IN DISHERGARH SEAM OF CHINAKURI NO.1 MINE

Scientific studies: Burst-proneness

- ✓ Burst-proneness was estimated from the energy index of the seam and the caving index of the roof rocks.
- ✓ The former was obtained was 4.02.
- ✓ The strata section was next studied (Sarkar, 1994) to give the results of Figure in which the caving index "I" was obtained as in the previous case of Parascole.
- ✓ Since difficulties in caving are indicated when **I > 2000**, these strata could be defined as hard.
- ✓ At German yardstick (Brauner, 1988) suggests that if a hard bed which is at least 5 m thick does not lie more than 19 m above the seam, it may cause bursts.
- ✓ The depth of 600 m, coupled with high values of energy and caving indices, indicated the Dishergarh seam to be burst-prone.

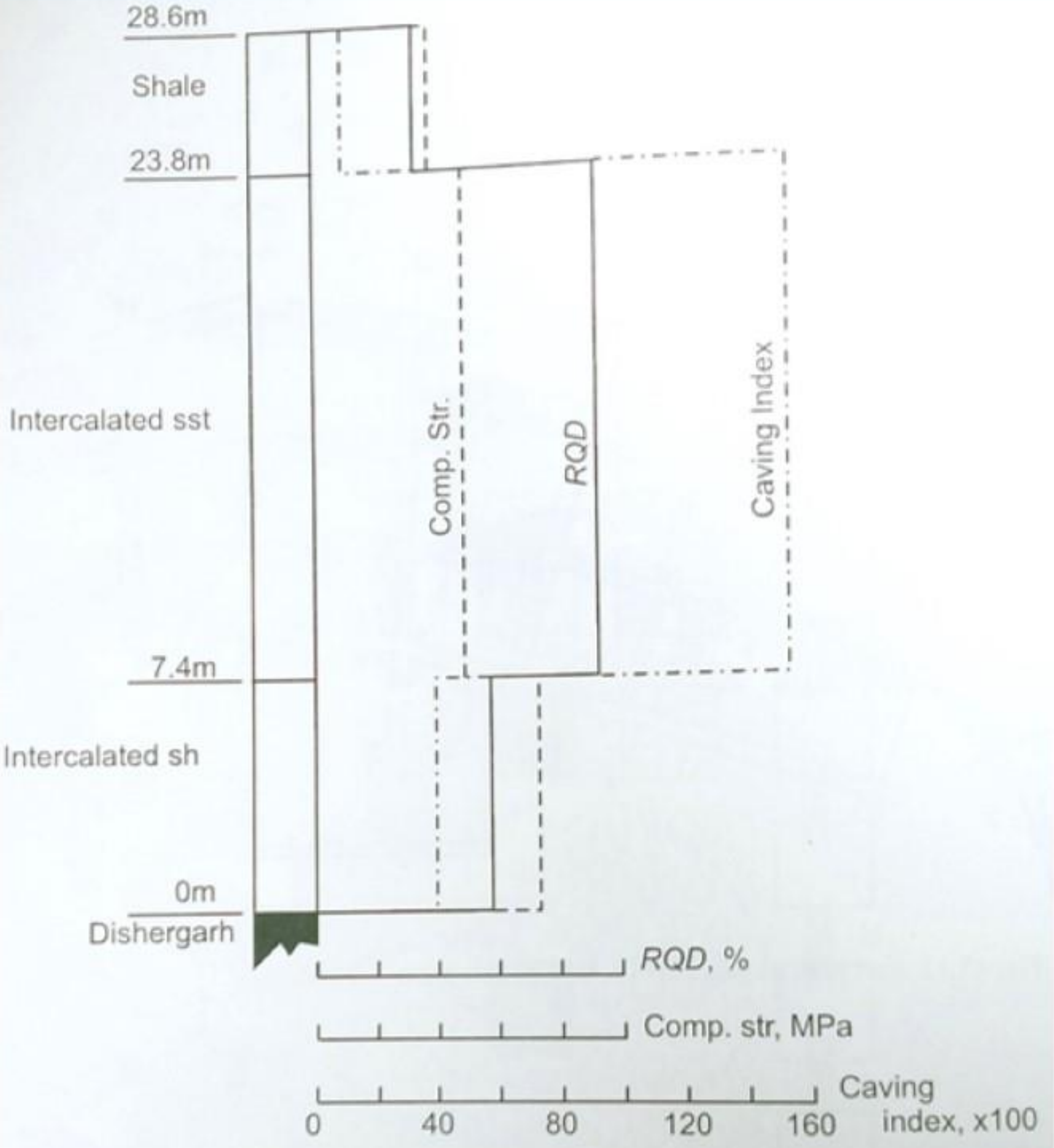


Fig. 13.13. Roof strata characteristics over Dishergarh Seam (after Sarkar, 1994).

ROCK BURST IN DISHERGARH SEAM OF CHINAKURI NO.1 MINE

Scientific studies: Field observations

- ✓ For drilling yield test a 42 mm diameter hole was drilled to a depth of 6 m (3 h, h being extraction height) with face advance.
- ✓ As per this test if gummings exceed 6 litres/m, a bump-prone situation exists.
- ✓ Figure shows the test results for 3 different rib widths.
- ✓ When the rib was 8.5 m wide, a rock burst was indicated by the test on 09-10-1992 and it occurred on 10-10-1992.
- ✓ Another interesting set of observations was obtained using simple strain bars grouted in the pillar ahead of the face.
- ✓ Vertical displacement, monitoring regularly with face movement, showed accelerated tensile displacement when the rib width became less than 10 m (Figure) culminating in a rock burst.

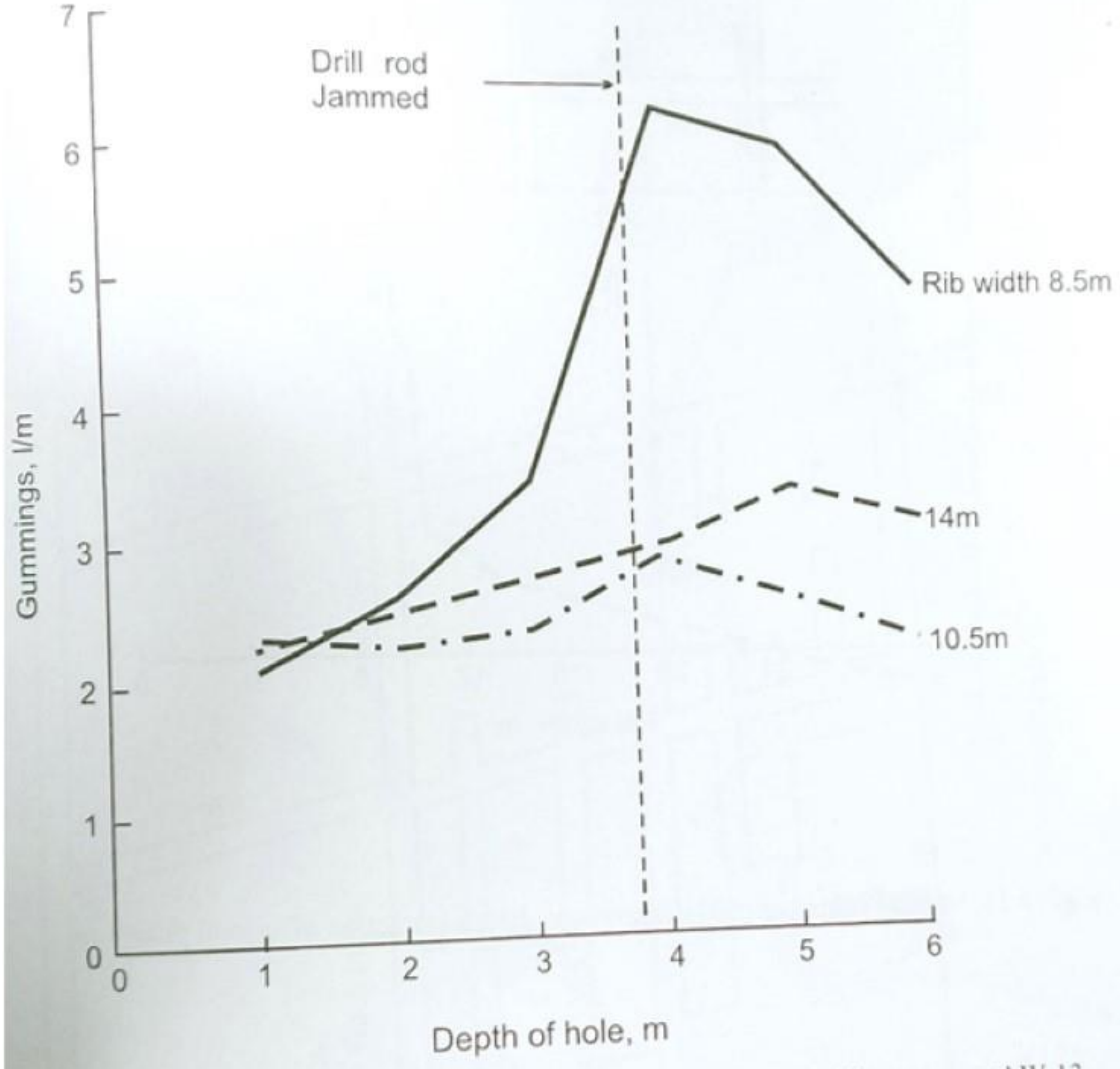


Fig. 13.14. Drilling yield test results during extraction of second pillar row, panel W-13.

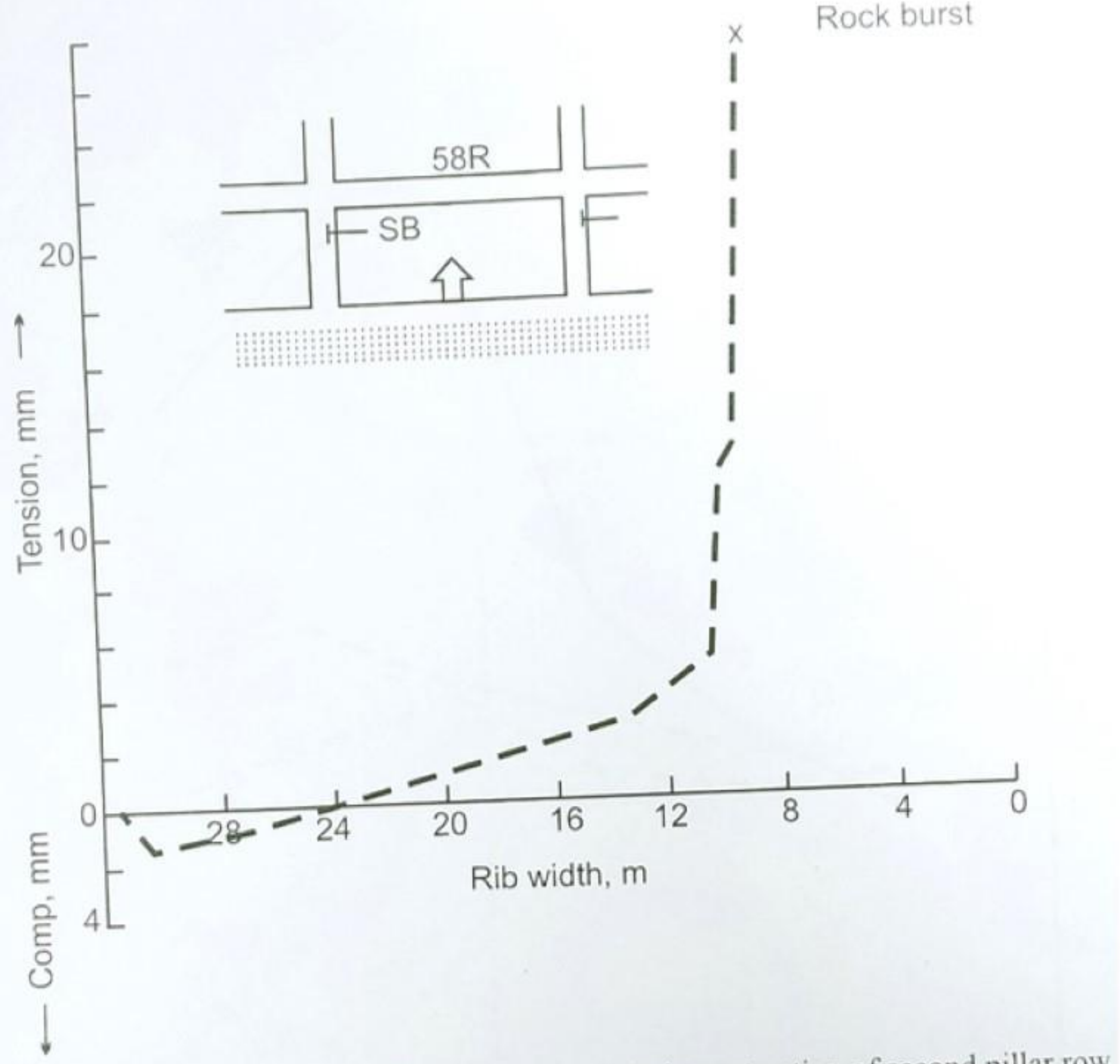


Fig. 13.15. Vertical strain bar (SB) observations during extraction of second pillar row.

CASE STUDY

ROCK BURST IN DISHERGARH SEAM OF CHINAKURI NO.1 MINE

Scientific studies: Prophylactic Measures

- ✓ From the experience in Polish mines (Goszcz, 1991) roof bursts occur when an element of the seam (e.g. remnant pillars) fails suddenly under high stress leading to sudden and violent shearing of the strata above.
- ✓ Roof bursts are a more dangerous phenomenon than coal bumps and the only relief measures appear to be good caving or solid goaf treatment.
- ✓ In the case of Chinakurl, since stowing was already being done, the only requirement was designing the ribs so that they remained stable till being buried in sand.
- ✓ The design of ribs was done using the BESOL 3D software employing the displacement discontinuity method.
- ✓ The BESOL model of the panel gave the average rib load as 28 MPa. The strength of the ribs was obtained CIMFR pillar strength formula as 27.1 MPa giving the safety factor as 0.97, indicating failure.
- ✓ From the above calculation and the strain bar observation (Fig. 13.15) the rib width was recommended as 10 m, which was increased to 12 min the DGMS permission.

CASE STUDY

ROCK BURST IN KOITHEE SEAM OF GIRMINT COLLIERY, ECL

- ✓ Panel 26-K of Girmint, where one serious injury occurred due to a coal bump on 19-02-1985, was probably the deepest case of depillaring with caving (depth 490m).
- ✓ It had the following particulars:

Seam	Koithee
Panel	26-K
Average depth	490 m
Extraction height	4.57 m
Seam thickness	4.57 m
Gradient	1 in 8.5
Pillar size between roadway centres	40 to 54 m
Roadway width	4.2 to 4.8m
Roof	Coaly shale

CASE STUDY

ROCK BURST IN KOITHEE SEAM OF GIRMINT COLLIERY, ECL

- ✓ The method of extraction consisted of conventional depillaring with ribs and slices and wooden supports.
- ✓ The site of the rock burst (bump), the position of the face etc. can be seen from Figure.
- ✓ Only a little caving had taken place until that date.
- ✓ Since CMRI was approached some time after the fact a physical picture of the incident was collected from the management.
- ✓ Considerable coal had dislodged from one side near the floor knocking a few props down, while the chocks remained standing.
- ✓ No roof fall at the accident site could be seen.
- ✓ This was thus a case of coal bump than of roof or floor burst.
- ✓ The initial DGMS permission allowed two dip-rise splits in a pillar with ribs 2.5 m and slices 3.9 m wide.

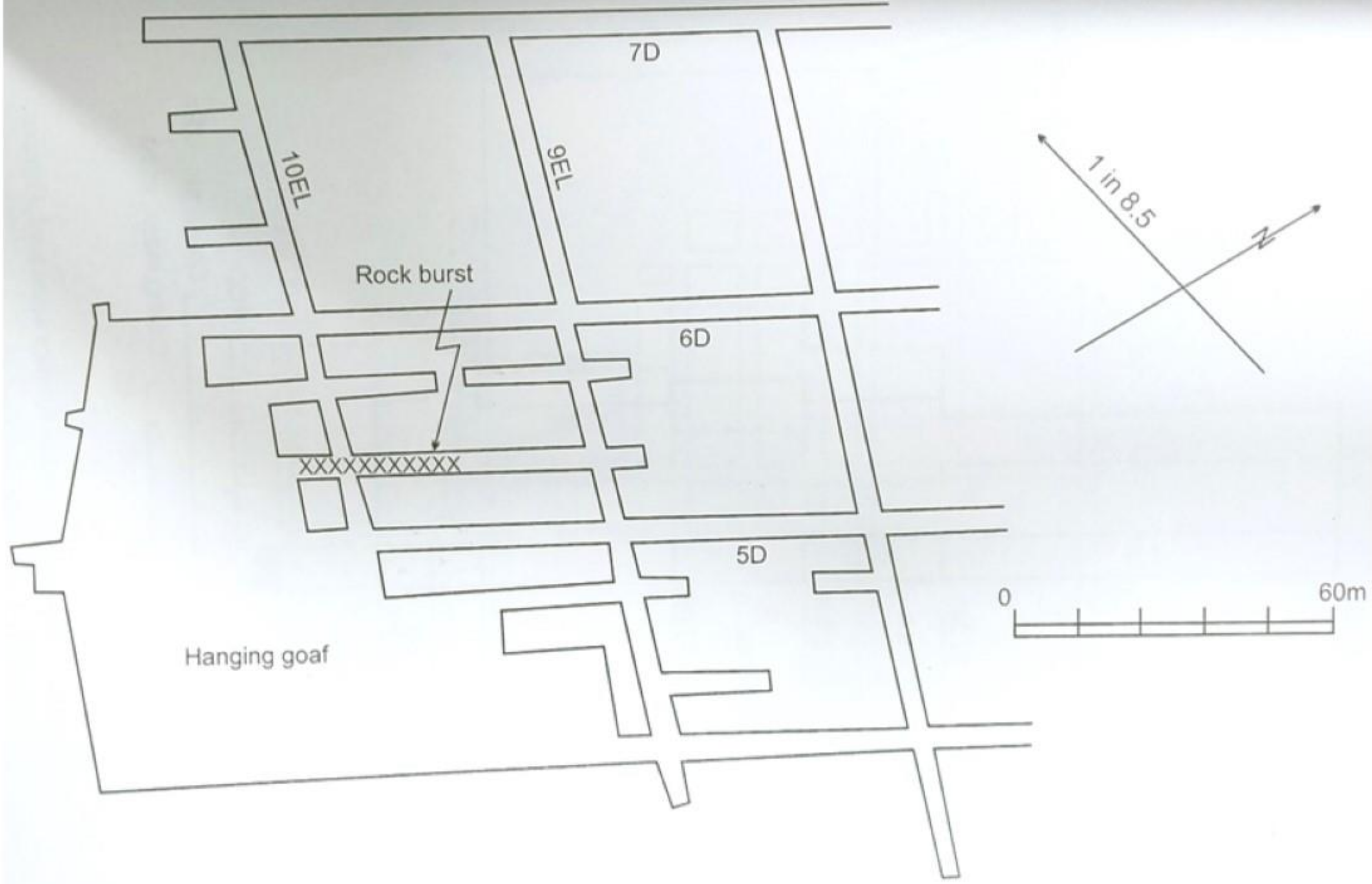


Fig. 13.16. Part plan of panel 26-K Koithee Seam, Girmint.

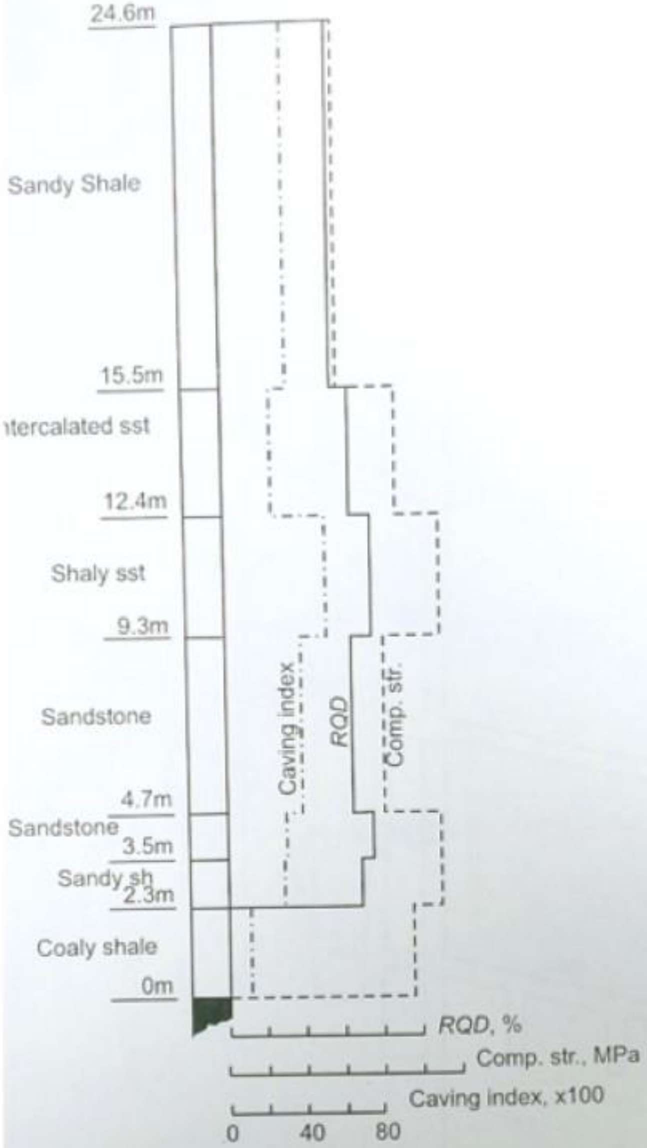


Fig. 13.17. Borehole section of Koithee seam roof, Girmint.

CASE STUDY

ROCK BURST IN KOITHEE SEAM OF GIRMINT COLLIERY, ECL

- ✓ After the accident the permission was revised to level splits (splitting being now restricted to only one row of pillars) and a slice width of 3 m only.
- ✓ CMRI investigations included determination of the energy index of the coal which was found to be 3.17, indicating moderate bump-proneness, and roof diamond drilling to obtain ROD and strength, which are shown plotted in Figure.
- ✓ CMRI also recommended side rope stitching.
- ✓ Further extractions in the panel could be done without incident.
- ✓ The BESOL model of panel 26-K is shown in Figure.
- ✓ A single 4 m slice was taken in the second step of the model to obtain Average ERR = 1.32x 10^5 J/m^2 Maximum ERR 1.50x10^5 J/m^2
- ✓ These ERR values are compared with Chinakuri conditions at the end of this case study. The vertical stress distribution over the 30 x 30 element window in Figure is shown in Figure. The CMRI caving index is plotted against each bed in Figure. These values indicate caving wit! difficulty. The conditions at Girmint in Koithee seam were thus burst-prone.

Comparison of ERR values

- ✓ Dishergarh seam at Chinakuri No.1 mine is of good quality, but is beset by rock burst problem and, naturally, cavability.
- ✓ In the interest of productivity and to eliminate the cost of stowing, longwall extraction with strong face supports was being considered.
- ✓ It was interesting to study the concept of Energy Release Rate (ERR) for longwall with caving as compared to-the prevalent depillaring wit stowing at Chinakuri as well as to the case of Girmint.
- ✓ For this purpose, several BESOL 3D models will various longwall face lengths were made, each model assuming sufficient face advance with artificially induced caving which would fill the goaf well.
- ✓ A comparison of the ERR values is shown in Figure, clearly indicating a high probability of rock bursts at Chinakuri if longwalling was adopted, in spite of good induced caving.
- ✓ As expected, the case of Girmint gives the least ERR, the depth of cover being an important factor.

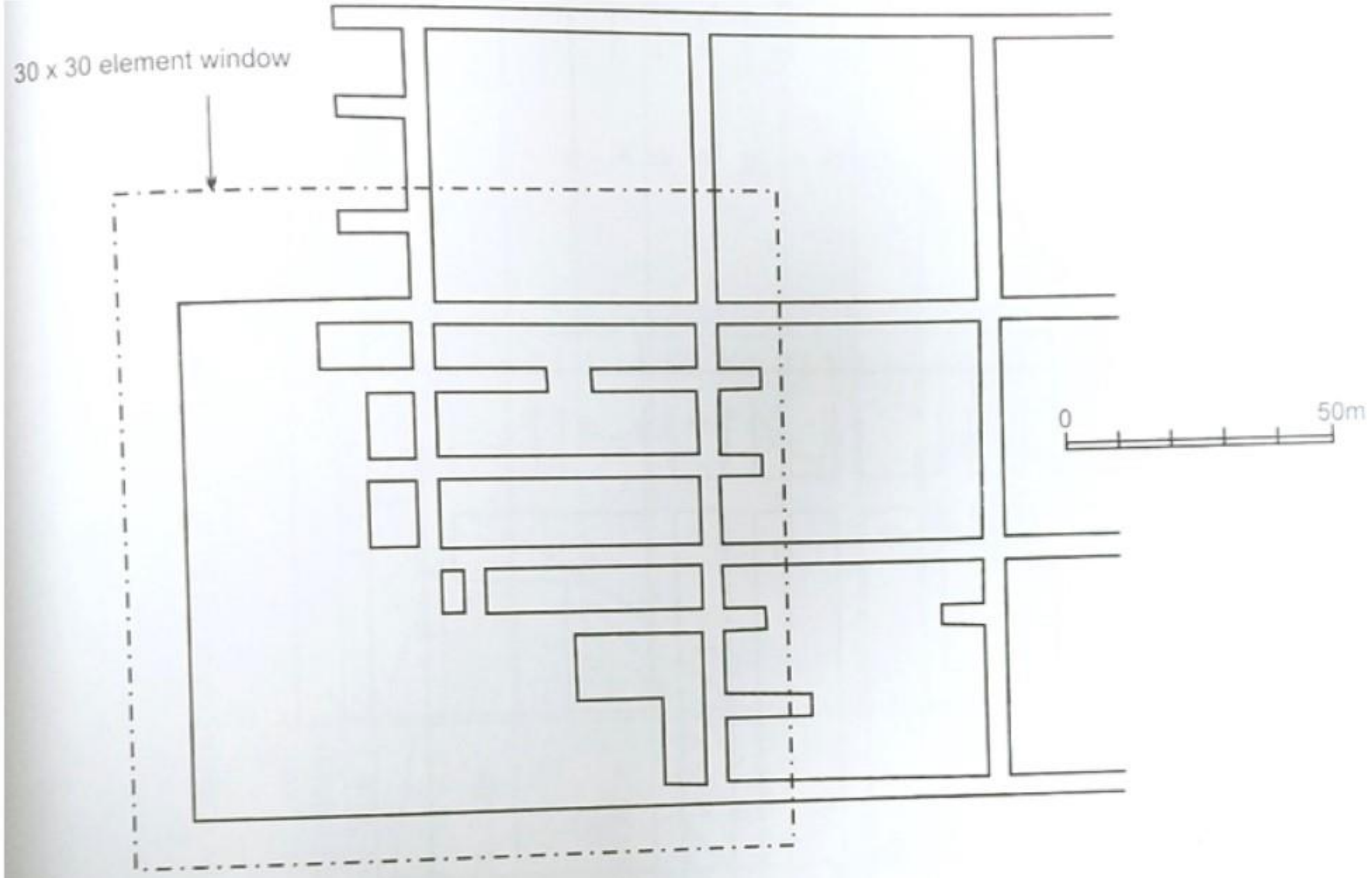


Fig. 13.18. BESOL Model of panel 26-K, Girmint.

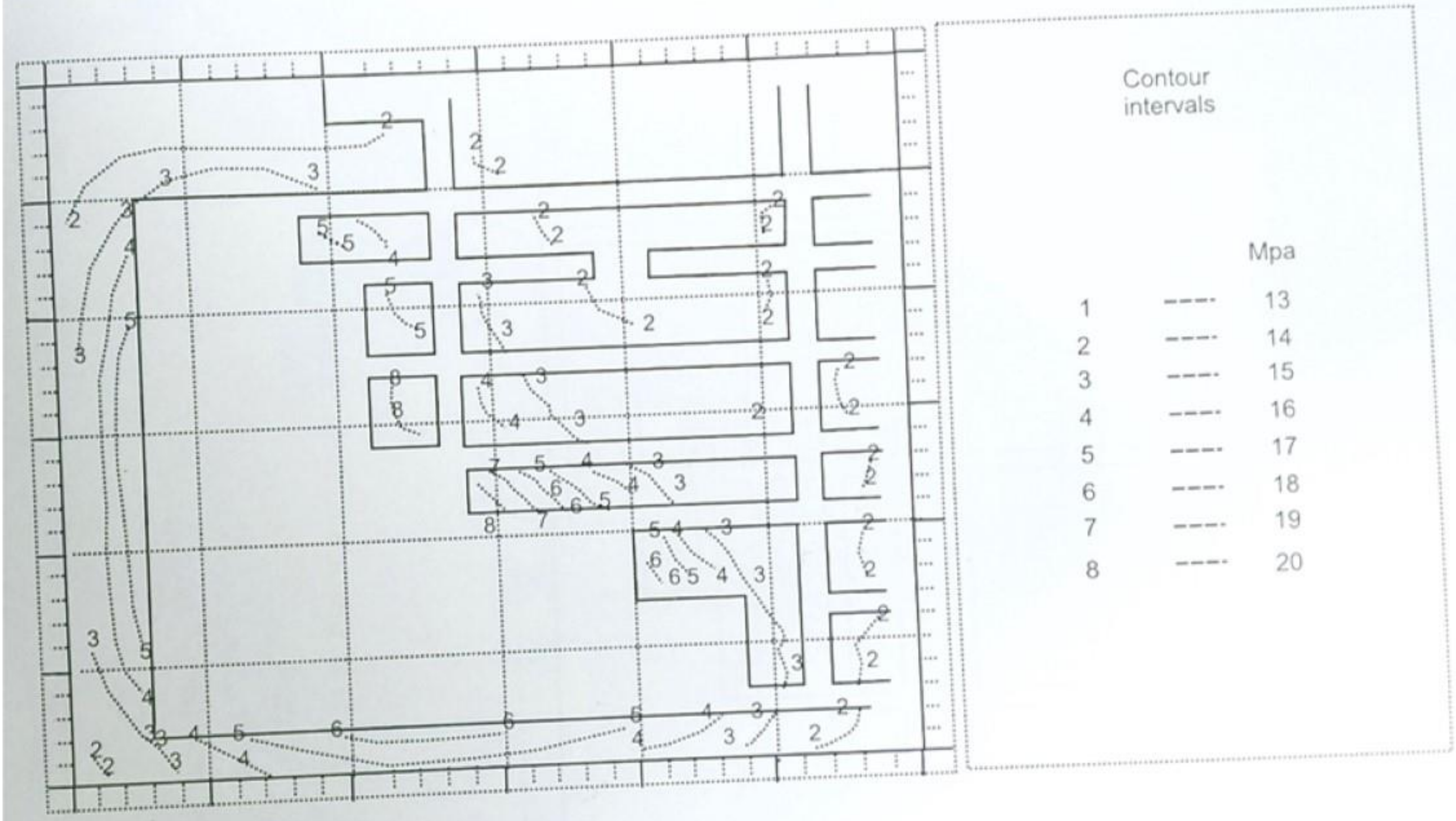


Fig. 13.19. Normal stress contours on panel 26-K, Kothee seam, Girmint.

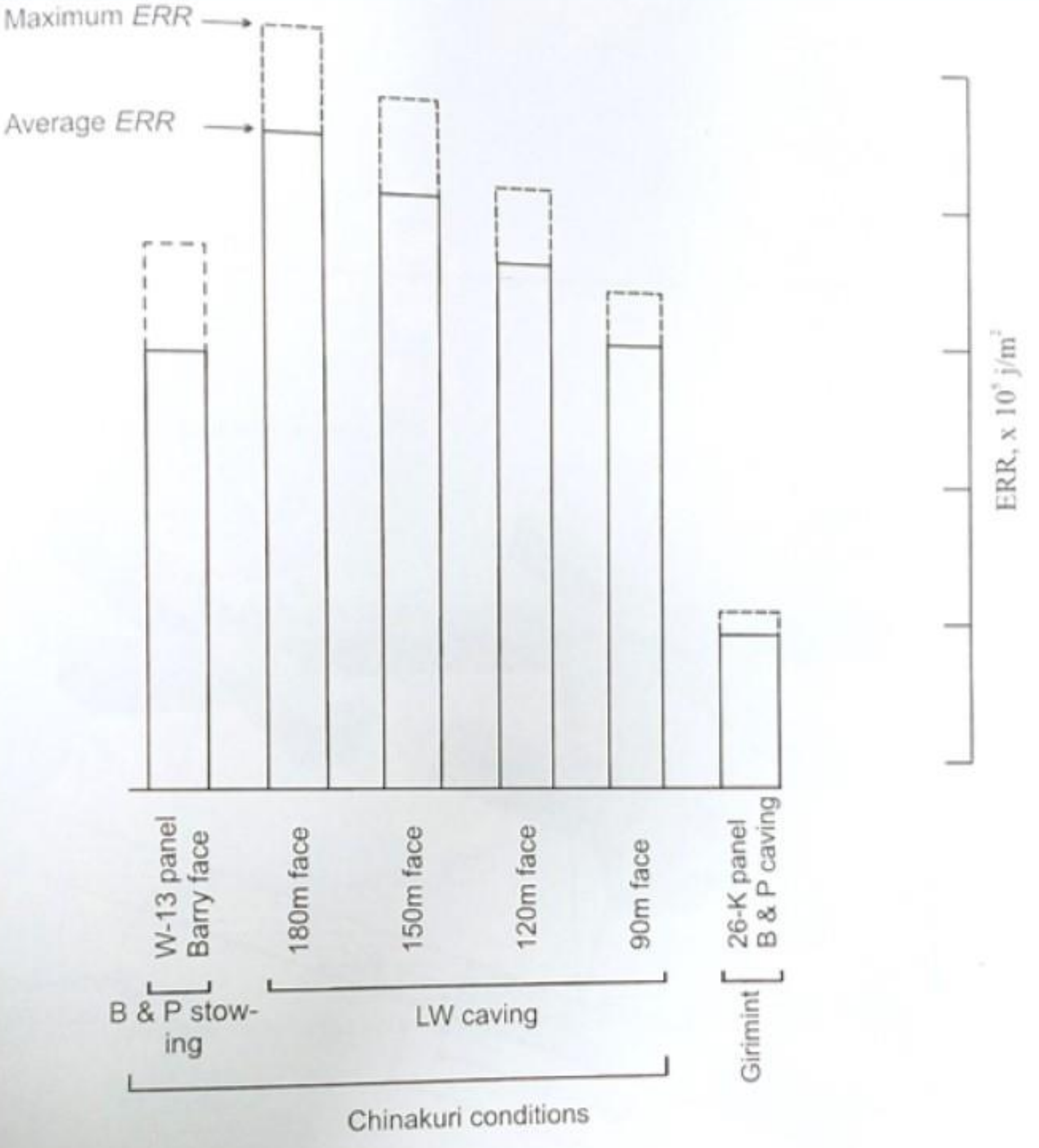


Fig. 13.20. Comparison of *ERR* values under Chinakuri and Girmint conditions.

Thank You

Queries??

Email: ashokmin@iitism.ac.in
Mobile: 8603401161